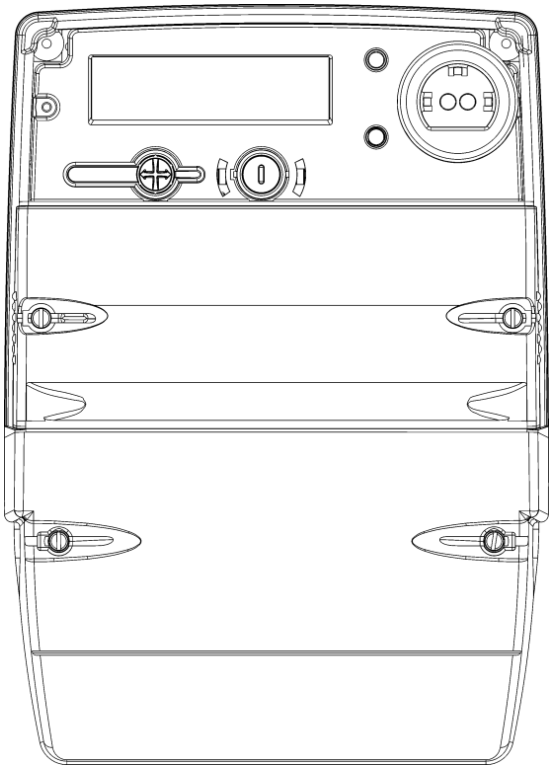
Premier 210 electricity meter with Optional RS-232/ NIC 5 communication module



User Manual

BGX701-075-R04

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As part of Secure Meter's continuous endeavour to improve product design, specifications mentioned are liable to change.

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Important

As part of Secure Meters continuous endeavour to improve product design, the specifications mentioned in this manual are liable to change anytime and therefore availability of features is product variant specific.

Specifications / features listed in this document are the most advanced available with Secure Meters on the date of release of this document, and hence should not be considered as default. Availability of all or some of the same depends upon the product variants and may not be readily available. Sales team should be contacted for queries, if any.

No part of the manual, or its content thereof, must be published, distributed, scanned or copied, in either electronic form or otherwise, without prior information and express consent of Secure Meters.

While all efforts have been made by Secure Meters to minimize errors, some errors may inadvertently exist. Secure Meters undertake to correct such errors wherever possible, and request feedback from users in this regard.

Secure Meters reserve the right to alter some or whole of the specifications mentioned in this document without any prior notice.

For any queries or clarifications, the user is requested to email the Secure Meters' support team at au.tech.support@securemeters.com

In most countries, electrical installations comply with more than one set of regulations issued by National Authorities or by recognized local or international bodies. Such regulations must be adhered to, even if they are not referred to directly in this document.

Secure Meters have duly conducted product quality tests as per regulations based on the observance of rigorous safety rules in the design and realization of the product.

In order to prevent tampering, the meter should be appropriately sealed after installation. Inappropriate handling of the seal may cause damage to the meter thus creating a false impression of being tampered with.

Please take note of the warning and caution icons presented in this manual as follows:

Possible Electric Hazard

This icon indicates the existence of dangerous electrical voltage. These operations must only be performed by qualified personnel.

Caution

This icon warns the user to take special precautions whilst performing an operation. The procedure must be followed as described in the manual.

Disclaimer

Secure Meters assume no responsibility for damage caused to the meter under the following circumstances:

- Irregular maintenance / improper installation
- Imprudence or carelessness during installation
- Normal (or abnormal) wear and tear of insulation
- Accidental contact with hazardous elements
- Immersion of meter in water
- Handling of meter by unauthorised persons
- Handling of meter by inebriated persons

Precautions and safety practices

- This product must be installed and serviced only by trained personnel. We strongly recommend reading the “Premier 210 installation manual” thoroughly before installing the product.
- Energy meters are generally installed in electrically hazardous areas. To minimise the risk of electrical shock, stay away from loose or exposed electrical connections. If there is loose or exposed electrical wiring near the meter, initiate appropriate preventive measures.
- The electrical installation of electronic meters at a site requires an adequate understanding of all governing rules and regulations.
- Keep the meter away from fire, rain and direct exposure to water.
- The meter may be damaged, either in part or completely, if it falls from a height.
- It is recommended to immediately cut off the electricity supply upon occurrence of a fault within the meter.
- Ensure that the load does not exceed the current range specified on the rating plate.
- Appropriate fuses and circuit breakers must be used in accordance with the rating of the meter.
- Secure Meters recommend regular installation checks to be carried out, especially when product is installed at locations posing high risk of degradation or corrosion, such as under direct strong sunlight, areas with high ambient temperature and little ventilation, areas near heat sources such as a furnace, areas where the meter is regularly exposed to chemical fumes, etc.
- When enabling the external antenna always ensure that it has been plugged into the SMA connector under the module.
- To ensure optimal performance of the radio system with the meter, the radio antenna should be fitted externally if the meter (or modem) is installed inside a metal enclosure.
- Always refer to installation & associated documentations and follow the instruction and the practices described. Manual should be strictly referred for details wherever the caution symbols are used on the product.
- If the meter is not used in the manner specified in the manual, the protection provided by the meter may be impaired.
- Safety practices must be followed as open position of the supply or load control switch does not provide isolation from the mains network.
- The installer is responsible for coordinating the rating and the characteristics of the supply side over current protection devices with the maximum current rating.
- Supply to the metering equipment must be isolated by removing active isolation link at the supply side to allow safe access before installation or removal. User is not allowed to connect any protection devices.
- Do not use solvents or abrasive materials to clean the meter, use only a soft and dry cloth and isolate the meter from the supply before cleaning it.
- Premier 210 is maintenance-free meter as it has no interchangeable or serviceable parts. To ensure a robust user experience, adhere to the environmental and technical specifications; and follow the instructions and safety measures to avoid risk of injury or damage to the equipment. For any further queries, you may contact the nearest sales representative.

Commonly used acronyms

Acronym	Description
ETBC	Extended Terminal Block Cover
ANSI	American National Standard Institute
RJ-45	Registered Jack 45
NMS	Network Management System
O/P	Output
NIC	Network Interface Card
PC-FR	Fire Retardant- Poly Carbonate
LCD	Liquid Crystal Display
P-N	Phase-Neutral
LED	Light Emitting Diode
SMA	Sub-Miniature version A
STOU	Seasonal Time Of Use
DIP	Demand Integration Period
PCB	Printed Circuit Board
LAN	Local Area Network
WAN	Wide Area Network
HAN	Home Area Network
HES	Head end system
ID	Identity
CIP	Consumer Interface Panel
TOU	Time Of Use
MD	Maximum demand
SIP	Survey Integration Period

Acronym	Description
QoS	Quality of Supply
CT	Current Transformer
IHD	In-Home Display
IEC	International Electro-technical Commission
BCS	Base Computer System
I/O	Input Output
UTRN	Unique transaction reference number
UART	Universal asynchronous receiver/ transmitter
AP	Access point
Eng	Engineering
RMR	Remote Meter Reading
DLMS	Device Language Message Specification
OBIS	Object Identification System

1 Product overview and operating principle

This manual is intended for understanding the various functions of Premier 210 and how to use its available features.

Broadly, the manual covers information about the available product builds, general technical specifications, details about the various applications and features, the configuration process both from the meter's Setup Mode (push buttons) and using M-Cubed, the various displays supported by the meter and Appendices on useful information.

1.1 About Premier 210

Premier 210 is a three phase, single element CT operated meter with fixed CT ratings. It supports credit mode of operation for Smart Metering. The meter supports 230/240V (+20%/– 20%) connection with the current of 5-20A. It has a relay of 2 Ampere and two fixed pulse outputs, one for Active Import and other for Reactive Import. It is used with optional RS-232/ NIC 5 communication module.

The communication module sits on front belly of the meter cover. It serves as the communication gateway to the Head End System (HES) via a Wide Area Network (WAN).

The modular communication design provides the flexibility to upgrade or change the meter's communication technology without the need to remove the meter or break metrological seals. The communication module connects the meter to HES over NIC 5 Mesh protocol. This communication provides remote data transfer between the meter and HES either as per the request or according to the schedule. The optional Zigbee module connects the communication module with other HAN devices –In Home Display (IHD).

It has an ANSI optical communication port for local communications and a user interface consisting of two push buttons which allows the user to select and display the data.

Commands or changes of configuration can be delivered to the meter in the form of authorised messages or UTRNs. UTRNs are encrypted codes that are valid only in the meter for which they are generated.

Relay disconnection / reconnection can be done remotely over the WAN or locally. A Mains frequency based clock or a battery-backed internal clock maintains the meter's date and time. Removal of the terminal cover and main cover is detected and logged in power-on condition.

Internal sensors monitor external magnetic fields allowing magnetic events to be logged and reported. The meter supports a maximum of two pulse outputs for calculating energy consumption. In addition, a 2A relay is integrated into the meter for demand management.

Note:

The operational scenario is similar for Skyline-i 033 except that the communication with IHDs over HAN is not supported. For Skyline- i 033 variant, any functionality, feature or step related to IHD or HAN must be ignored throughout the manual.

1.2 Premier 210 Connection Terminals

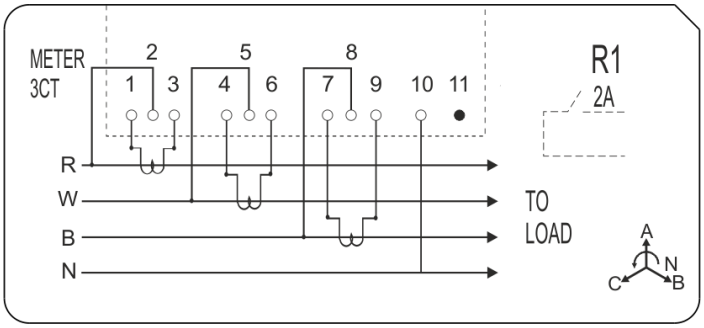


Figure 1: Internal Connection for Premier 210

1.3 Premier 210 Technical Specifications

Table 1: Technical s

S. No.	Specification	Description	
1	Mechanical	Dimensions (mm)	252 (H) x 180 (W) x 118 (D) without hook (with 35 mm extended main terminal cover)
		Weight	2.2 kg approx. (Actual weight may vary with variants)
		Mounting	Projection type
		Front cover	Translucent PC-FR grade V0
		Base	Opaque, PC-FR material
		Communication module cover	Opaque, PC-FR material
		Terminal cover	Opaque, PC-FR material
		Secondary terminal block cover	Opaque, PC-FR material
		Extended terminal block cover	PC-FR grade V0, Opaque without U cut, 35 mm from terminals
		Terminal block	Opaque 10% glass filled PC-FR grade V0, matt finish
		Rating plate	Opaque, PC-FR material
		Main load terminal screws	Screw M4*8mm (brass, nickel plated)

		Minimum bore diameter for main terminals	5.0 mm, suitable for 4 mm^2 multi-strand copper cables
		Bore Diameter of Pulse I/O	4 mm, Suitable for 2.5 mm^2 cables
		Sealing provision	• 2 seals at the back of the meter (upper screws) • 2 seals under the terminal cover • 2 seals on the terminal cover • 1 optional seal for configuration button • 2 seals on the module
		Sealing screw	Captive type, made of brass, sealing hole diameter 2mm
		Display	Super-twisted Nematic (STN) LCD with backlight
2	Electrical	Connection, Supply	3 Phase, 4 Wire, Current transformer operated
		Voltage measuring element	Voltage divider
		Current measuring element	CT
		Measuring Elements	Three
		Reference Voltage (Supply Voltage)	230 / 240V± 20%
		Reference current range	5A Ib, 20A Imax (per phase)
		Frequency	50 Hz ±5%
		Power Factor	0.5 Lag - Unity - 0.8 Lead as per IEC/AS 62053-21 for Active Energy 0.25 Lag - Unity - 0.25 Lead as per IEC/AS 62053-24 for Reactive Energy
		Accuracy	Class .5 for Active Energy Class 2.0 for Reactive Energy
		Metrology LED	Two LEDs, 5000 pulses/unit
		RJ45 port/connector	Port accessible (RS 232)
		Load control relays	230/240 VAC, 2 A (volt free)

			Type: PVC, V-75, Copper Size: 1 mm^2 Voltage rating: 600/1000 V Normal temperature rating: 75°C	
		Pulse Output	27Vdc, 27mA (volt free) Type: PVC, V-75, Copper Size: 1 mm^2 Voltage rating: 600/1000 V Normal temperature rating: 75°C	
		Non-volatile memory	8 MB Flash, Data retention up to 15 years	
		Connecting Cables (Recommended -As per service and installation practices)	Type	PVC, V-90, Copper
			Size	4 mm^2 (conductor diameter) for 20A
			Voltage rating	600/1000V
			Normal temperature rating	75°C
		Communication module	Nominal Voltage: 4V(+/-0.1V) Maximum load: 1.4A (Peak)	
3	Protection	Dielectric Strength	• Withstands 415 VAC continuous • Impulse withstanding as per AS and NMI M6	
4	Compliance	Reference Standard	AS62052.11, AS62053.21, AS62056.21, AS62054.21, AS62053.24, NMI M6-1, AS62052.31	
		Lighting Interference	AS62056.21 (16k Lux under controlled conditions)	
		Environmental Protection	IP 54 with suction	
5	Environmental	Storage Area	The meter is designed for indoor (/outdoor)* use, in accordance with IEC 60721-3-3:1996. *Ideally the meter should be stored and operated in a dry, well ventilated, climate-controlled building. Rapid changes in temperature and humidity should be avoided.	
		Operating Temperature as per AS standard	**Range**	**Outdoor Meter**
			Specified operating range	-10 °C to +55 °C

			Limit range of operation	-10 °C to +70 °C
			Limit range for storage	-25 °C to +70 °C
			Limit range for transportation	-25 °C to +70 °C
		Operating Temperature as per NMI M6	Specified operating range	-10 °C to +60 °C
			Limit range of operation	-20 °C to +70 °C
			Limit range for storage	-25 °C to +70 °C
			Limit range for transportation	-25 °C to +70 °C
		Operating Humidity	Annual mean	< 75 %
			For 30 days, spread in a natural manner over one year	95 % (non- condensing)
			Occasionally on other days	85 %
6	Power Consumption (As per IEC 62053-61)	Voltage Circuit	< 2 W/phase and < 3.3 VA /Phase average Skyline-i 033 fitted) < 2.5 W/Phase and < 3.5 VA/ Phase (with communication)	
		Current Circuit	< 0.2 VA @ Ib/phase and < 7.5 VA @ Imax/phase as per IEC 62053-61	
7	Additional information	Altitude level	Altitude level up to 2000 m from sea level	
		Meter type	Outdoor	

1.4 Handling, storage and operating conditions

The Premier 210 meter is an electronic device containing delicate components, which should be handled carefully during transit, storage, and installation. The meter should be protected from physical vibration and shocks. Wherever possible the meter should be kept in its original packaging until it is installed at the customer's site. Temperature and humidity must be maintained within the limits expressed in the above table.

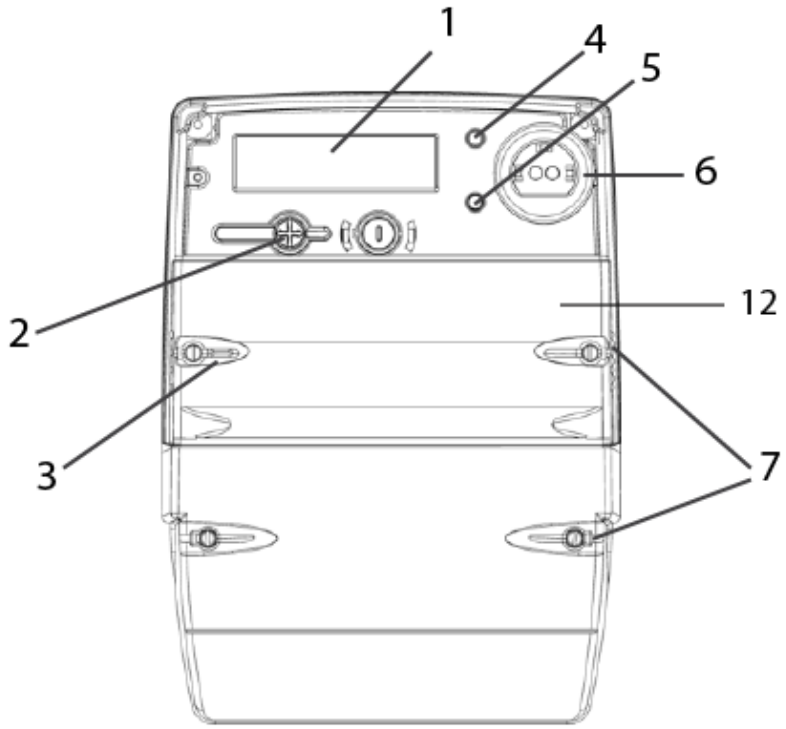
Proper installation and removal procedures should be followed, in order to prevent damage or injury. Physical damage to the meter's case could indicate damage to internal components. Under no circumstances should an attempt be made to install a damaged meter.

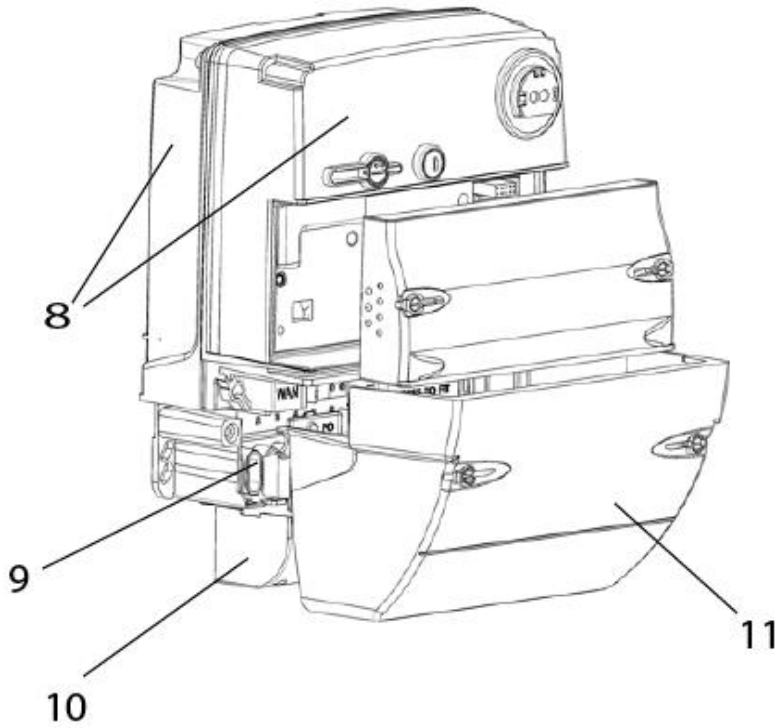
A visual inspection should be performed before installing the meter. The following should be checked, as a minimum:

- No evidence of external damage or missing parts
- No missing or damaged wiring
- No evidence of overheating

Ideally the meter should be stored and operated in a dry, well ventilated, climate-controlled building. Rapid changes in temperature and humidity should be avoided.

1.5 Premier 210 Parts





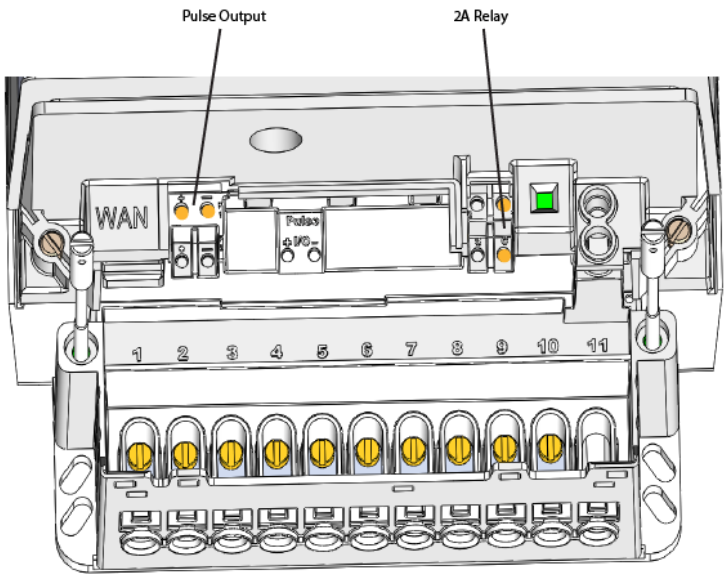


Figure 2: Premier 210 Part Details

Fixed Plate–Information is fixed and is common for all meters of this type.

1. **LCD display -** A large multi-segment LCD supporting alpha-numeric characters displays the information and values for various electrical parameters recorded by the meter.
2. **Push Buttons -** Two push buttons Left and Right, allow interactions with Premier 210. These buttons are used for various user operations such as display navigation, boost operation etc.
3. **Communication Module –** To facilitate AMI communication, Premier 210 provides a modular compartment; into which, a communication module can be fitted as required.
4. **Metrological LED (Fixed) -** This LED is provided on the meter main face plate and flashes according to the rate of energy consumption measured by the meter. It is used for testing the accuracy of the meter and as a visual indication of the rate of energy consumption.
5. **Metrological LED (Configurable) -** This LED is provided on the meter main face plate and is configurable at production for any of the four Quadrants.
6. **ANSI optical port -** An electrically isolated optical port is provided for external communication to prevent tampering / damage to the meter from spurious external electrical signals.
7. **Sealable Screws -** These screws facilitate the sealing arrangement necessary after the installation of the meter in the field.
8. **Meter Base and Front Cover -** The base or enclosure of the meter is made of a high-grade fire-retardant poly-carbonate. The front cover, made of the same material, is fastened to the base with screws. A sealing arrangement is provided to prevent tampering.
9. **Terminals and Terminal Block -** A terminal block, capable of sustaining high temperature, is provided in the meter with adequate provisions for connecting wires to the meter.
10. **Secondary Terminal Cover-** The secondary terminal cover provides additional protection for the live terminals of the meter and keeps them inaccessible even after removal of the main terminal cover.
11. **Extended Terminal Cover–**The extended terminal cover protects the terminal connections from tamper or interference, provides safety from hazards and provides security of add-on modules**.**
12. **SMA connector for external antennas -** SMA (Sub-Miniature Version A) connector is available under the module to connect a compatible external antenna (WAN and HAN) on the Premier 210.

13. **RS-232 Interface** – An optional RS232 interface for plug-in devices on an RJ45 port is provided for remote communication. This interface can be used as an alternative for remote communication when the NIC module is not being used.
14. **Pulse Output**– Pulse outputs are generally used to indicate a variety of available events or states. The meter has one 27V DC@27mA pulse output. The output will be potential free.
15. A 2A Relay is provided in the Premier 210 meter which is used for managing load control applications.

2 Premier 210 display and button details

Premier 210 is equipped with a large multi-segment alpha-numeric LCD display and two buttons for the essential user interface.

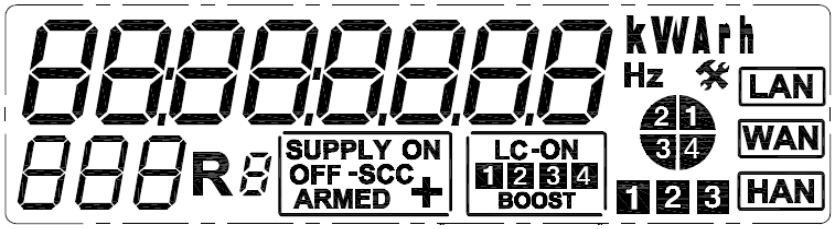


Figure 3: Segment Check for the LCD Display

The LCD screen has various sections and annunciators for displaying specific bits of information such as relay or load connection status, parameter values, user messages etc. An annunciator or a section corresponding to the demanded information is illuminated at the time of display. When powered, the LCD is capable of displaying a group of related electrical parameter values together on a single page.

Details of individual sections are summarised below:

Table 2: Premier 210 LCD Segment / Annunciator Details

S. No.	Icon / Segment	Details / Remark
1.	88888888	Alpha-numeric main value indicator - This segment displays the values of pre-configured parameters in an interactive mode. It also displays the health status of the meter and messages on the transactions performed with the meter.
2.	kVArh	Unit indicators - These icons display unit values using combination of characters like k, V, W, A, r and h.
3.	+	Sensitive load indicator -This icon indicates presence of Sensitive load mode in the meter.
4.	Hz	Frequency indicator - This icon indicates supply frequency unit.
5.	888	Display code indicator - This icon indicates individual display ID assigned to each of the displayed parameter. This code allows distinction between two similar displays.
6.	R8	NOT USED
7.	2 1 3 4	Energy quadrant annunciators - glowing of a particular quadrant indicates the instantaneous information about the energy flow, as follows - 1 Active import, Reactive import, Lagging PF 2 Active export, Reactive import, Leading PF 3 Active export, Reactive export, Lagging 4 Active import, Reactive export, Leading PF
8.	LC-ON 1 2 3 4 BOOST	LC-ON 1 represents the relay state (ON/ARMED).

9.	SUPPLY ON OFF-SCC ARMED +	Supply will be permanently in ON state SCC: Not used
10.	BOOST	NOT USED
11.	LAN	NOT USED
12.		Service status mode indicator - Indicates that attention and due service by authorised personnel is needed for maintenance e.g. Real Time Clock failure.
13.	1 2 3	Phase annunciators - Indicate the presence of supply voltages at the respective phase. The annunciators also flash if reverse current is detected in any of the phase.

Table 1: Operations possible through Premier 210 push buttons

S. No.	Function	Details
1.	Parameter Navigation	Premier 210 may be configured to display a desired list of parameters. These parameters are logically grouped to appear as a different display page, each of which is assigned an individual display ID.
2.	Metrological LED configuration	The metrological LED can be configured to flash on desired energy channel using the parameter navigation function. **Note:** This is applicable only till midnight cross over, following which, it switches to the factory configured settings. The same is applicable for a power interruption.

2.1 Using the display

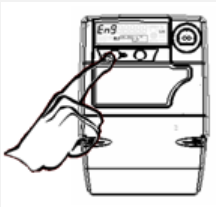
Two modes are available for viewing the displays in the meter:

- Push Button Mode
- Auto Mode

Push Button Mode

The push buttons provided on the front panel can be used to navigate through various displays, switching to a different display mode.

Table 2: Push Button Display Mode

S. No	Button	Action
	Left button	Advances to next page or display the first parameter of the next page.
	Right button	Selects the next parameter in the current page sequence

Auto Mode

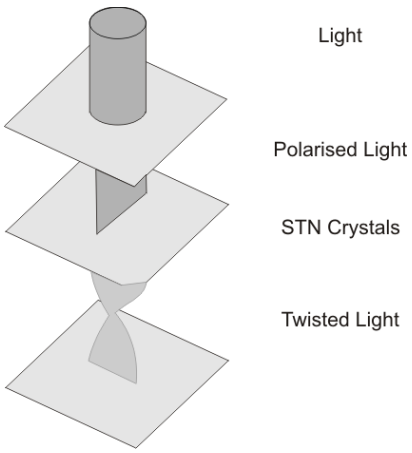
Auto Mode is the default mode for Premier 210 display. The configured parameters scroll automatically within a specified period. The auto cycle ends with a brief sleep mode of two minutes, following which the display cycle is repeated. Upon power interruption, the display starts in auto mode.

A complete list of sample display screens are shown under **Error! Reference source not found.**.

2.1.1 Display types and viewing arrangements

Premier 210 uses a Super Twisted Nematic (STN) type Liquid Crystal Display (LCD) which offers more contrast than normal TN types. Twisted Nematic Display is the most commonly used crystal display used in digital devices. A STN display contains a liquid crystal sandwiched between two plates of glass, which twists and un-twists at varying degrees to allow light to pass through. When no voltage is applied to a STN liquid crystal cell, the light is polarised to pass through the cell. In proportion to the voltage applied, the liquid crystal cells can twist from 180° to 270° changing the polarisation and blocking the light's path.

The example shown below illustrates how the longitudinal polarisation is twisted after passing through STN Crystal Display.



The figure below illustrates the meter display from two different positions:

Case 1: When the meter display is viewed from a level above the meter.

Case 2: When the meter display is viewed from a level below the meter.

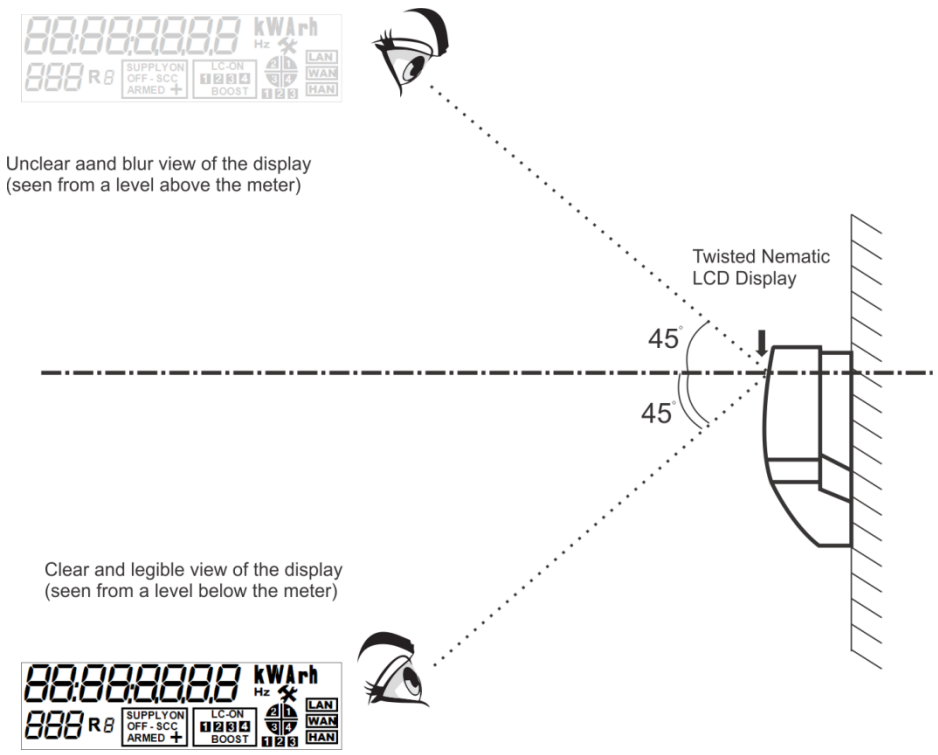


Figure 4: LCD Viewing Angle

3 Meter Configuration

A configuration is a pre-defined set of rules used to define the meter functions and tariff. Premier 210 can store multiple configurations (upto 16) and is factory configured with either a default configuration or a customer-specific configuration. However, it may be configured anytime at the customer site using the following tools:

- M-Cubed for configuring various firmware features. Configurations prepared using this tool are saved as files using extension 'xml', e.g. <filename.xml>
- Download Tool for downloading configuration file (*.xml file) into the meter - This operation is supported by M-Cubed and NMS. It is possible to download the configuration using both 'Local' as well as 'Remote' communications. One may use appropriate tools according to the communication setup available.

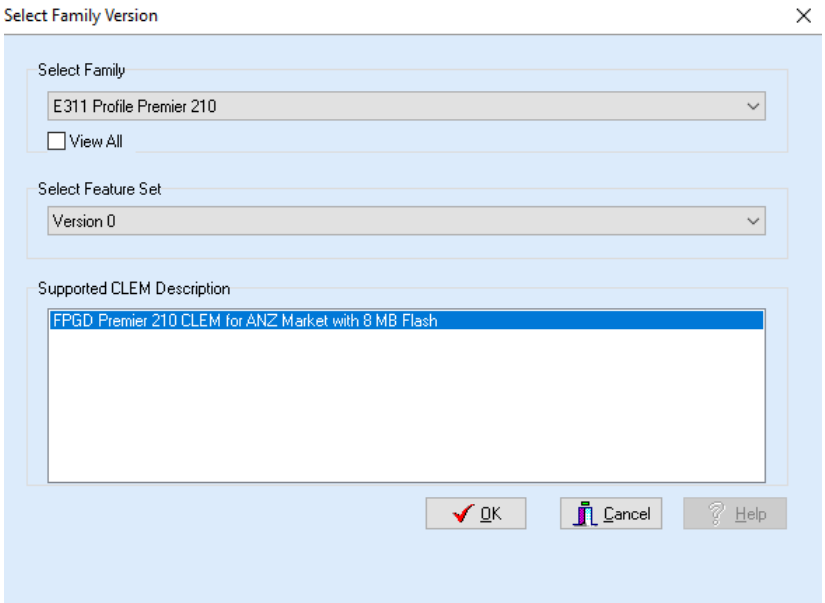
In this section you will learn about:

- Creating a configuration file under 3.1
- Creating a factory configuration file under **Error! Reference source not found.**
- Downloading a configuration file under 3.2
- Activating a configuration under 3.3

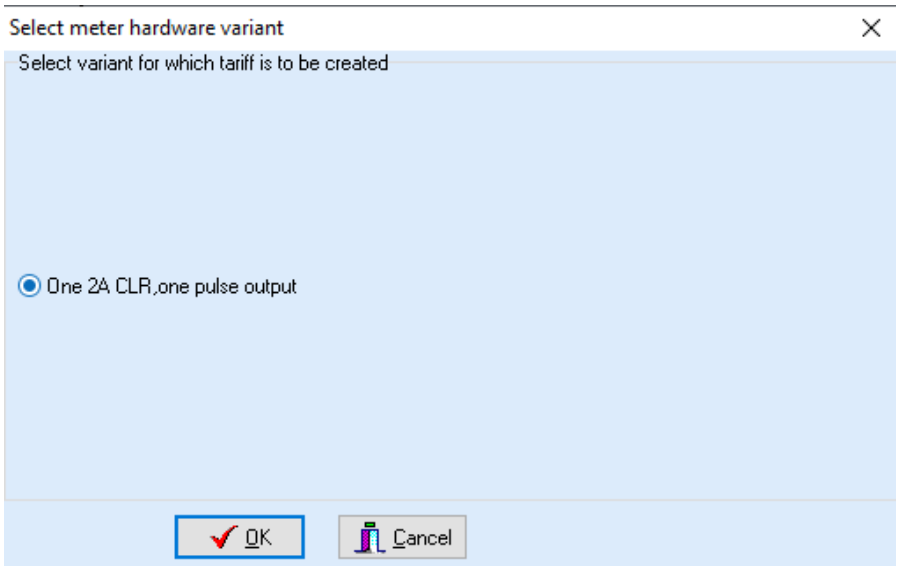
3.1 Creating a Configuration File

Premier 210 allows creation and storage of upto 16 configuration files. This allows for comprehensive tariff structure implementation to meet dynamic energy and rate demands. Configuration files are created using M-Cubed. This tool has a wizard-based interface which automatically provides instructions for creating configuration files in accordance with the selected CLEM type and hardware variant.

Step 1: In the wizard, select the correct firmware family, feature set for creating the configuration file and the supported firmware type. In the example shown below, fields are selected as per requirement.



Step 2: Select a variant type for which tariff file needs to be created. Subsequent screens will be based on the selected variant.



Step 3: Click the **OK** button to create the configuration file.

3.2 Downloading a Configuration File

Tools such as M-Cubed can be used for downloading the configuration file (*.xml file) into the meter. Any of the communication options described in the Communication section can be used for the same.

Configuration can be downloaded in the meter either locally or remotely depending on the available communication infrastructure. For local downloading, the following resources are required:

- M-Cubed Software
- ANSI optical communication cable

NOTE:

Using the same tools, it is also possible to upload/read-back the configuration available in the meter memory.

3.3 Activating the Configuration

There are two different ways to activate a configuration -

- Activation through Command Mode
- Activation through Engineering Mode

Table 3: Configuration Activation Methods

S. No.	Method	Details
1.	Command Mode	After the configuration(s) have been downloaded into the meter, it is easy to select and activate the required configuration by issuing a command from compatible software such as the M-Cubed (locally) or NMS (remotely). Remote activation would require an active communication link between the meter and the NMS.
2.	Engineering Mode	This method is used to activate a configuration manually at the site using the display and push buttons operation. Only stored tariffs can be activated using this method.

4 Configuring Premier 210 for AMI

A modular design and flexible technology allows Premier 210 to be programmed for catering to advanced metering needs such as TOU based rate register switching, Remote firmware upgrade, remote and local connection and disconnection, pulse output configuration, multi-channel communication support and dedicated event logs.

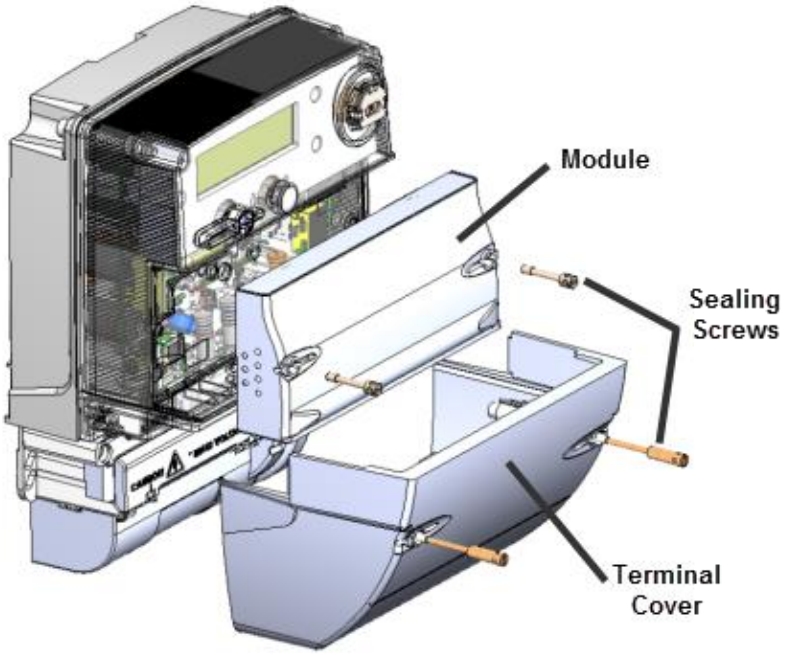
This section will take you through the basics of AMI and further configuring Premier 210 for the required Distribution Network usage:

- A brief on AMI under **Error! Reference source not found.**
- The AMI product offering by Secure under **Error! Reference source not found.**
- Add-On Modules details for AMI under 4.1
- External communication details for AMI under 4.2
- Mesh Radio communication details for AMI under 4.3
- WAN Port details for AMI communication under 4.4

4.1 Add-On Module for AMI

Add-On modules are accessories which can be attached to the meter for fulfilling special requirements.

Premier 210 supports different types of communication modules which can be fitted to the mid-section of the meter or placed under a terminal cover and connected to the RJ-45 socket on the meter.



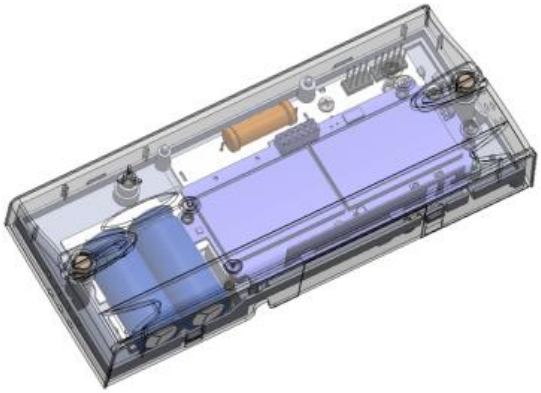


Figure 1: Premier 210 Communication Module

The rear view of a module is shown below. The module has location pins to stop inserting the module upside down. The example shown has a SMA antenna connector for fitting an external antenna. Be sure to feed the antenna cable through the edge slot. The module is held in position by the captive fastening screws. These also act as sealing points if required.

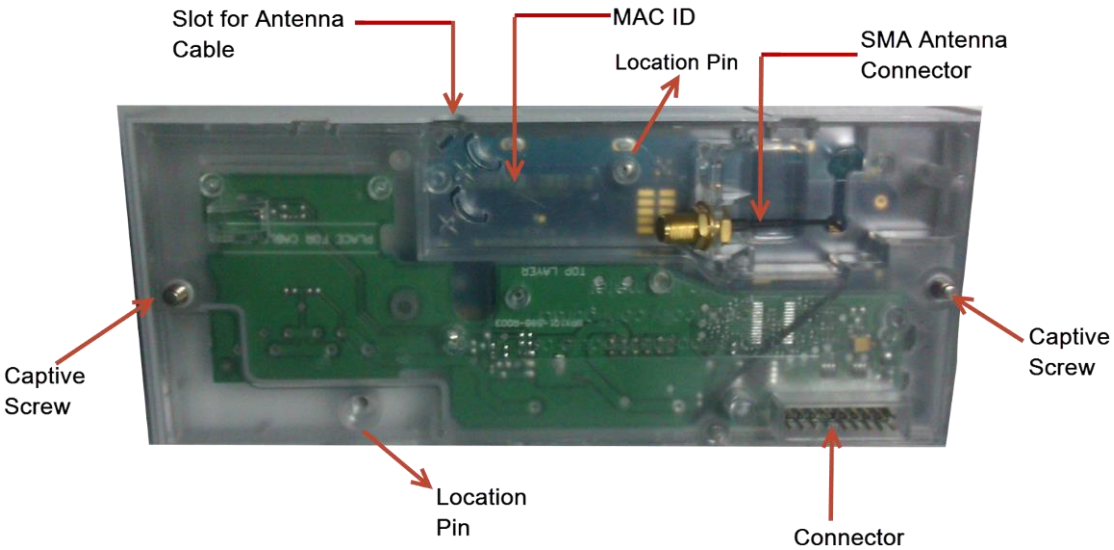


Figure 2: Premier 210 Communication Module Details

4.2 External Communication for AMI

Communication with the meter is necessary for performing various transactions such as meter reading, meter configuration, load control, firmware upgrade, etc.

Premier 210 supports following AMI communication modes:

1. Remote: Through Integrated RF mesh and Zigbee with module NIC - 512
2. Remote: WAN port (RJ-45) for communication when NIC-512 is not used

4.3 Mesh Radio Communication

An AMI module is integrated in the Premier 210 to communicate with the Access Point, AP over radio. This AP is further connected to NMS.

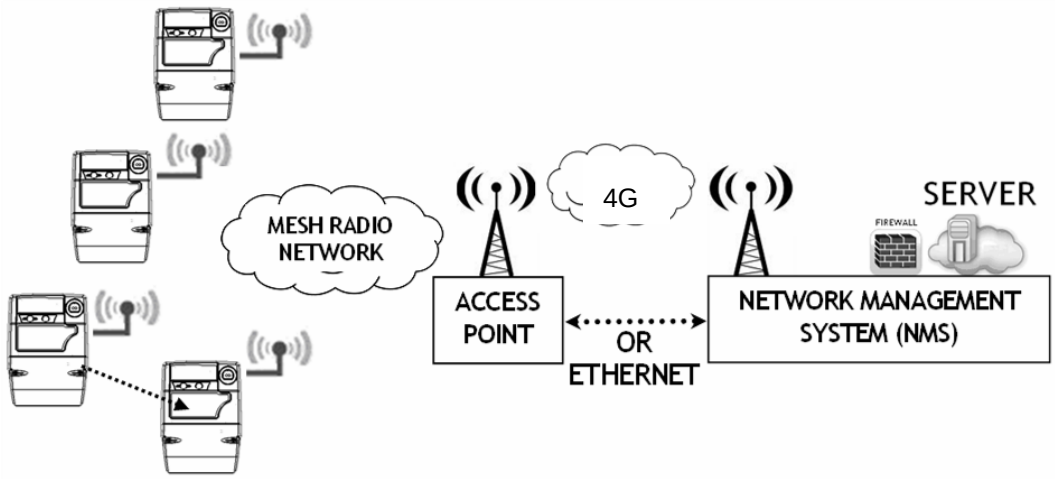


Figure 3: Mesh Radio Communication

4.3.1 Network Status and Control

The AMI module sends the status of the LAN, WAN and HAN networks to the meter. The meter further displays this status through appropriate annunciators as shown in the table below.

Table 4: Network Status and Control

Case	Status	Icon Status	Code
1.	Not available	OFF	0
2.	Access disabled	OFF	1
3.	Initialized	Flashing fast	2
4.	Found	Flashing slow	16
5.	Connected	ON	32

You can configure LAN, WAN and HAN settings in M-Cubed under **Transactions > Change Operations > LAN/WAN/HAN Configuration.**

The meter logs the network status at each minute boundary. If the status is not updated for a pre-defined period of time, the meter logs an event. The event is restored as soon as the network status is updated by the communication module.

4.4 WAN Port

The pin out details for the AMI module socket is shown below.

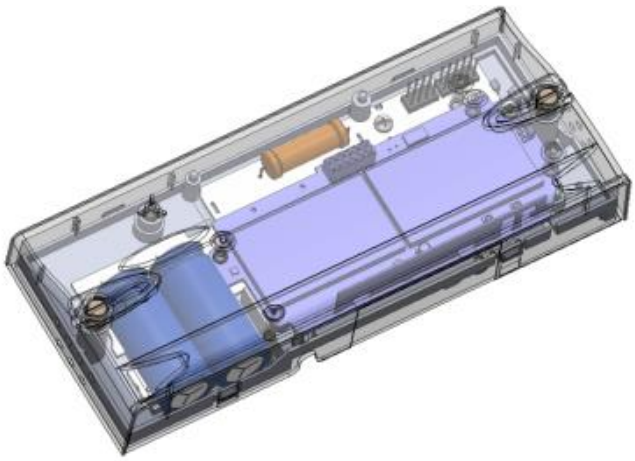


Figure 4: AMI Module Socket Pin Details

Table 5: AMI Module Socket Pin Details

NIC Signal	Pin No. for AMI module Connector	Description
NC	1	NC
GPIO	2	NC
GPIO	3	NC
VCC(4V)	4	DC 4 Volts -/+5% to deliver a maximum current of 2200 mA with voltage and current measured using a True RMS measuring instrument.
GPIO	5	NC
GND	6	Isolated Ground
TXD	7	Meter receives / AMI Module transmits. Interface default data rate shall be 9600 bps, 8N1. Interface shall use TTL signal levels (i.e. 3.3v). Interface shall communicate with the meter via PRIA's PACT protocol/DLMS protocol as agreed with SSN.
GPIO	8	NC

NIC Signal	Pin No. for AMI module Connector	Description
RXD	9	Meter transmits / AMI module receives. Interface default data rate shall be 9600 bps, 8N1. Interface shall use TTL signal levels (i.e. 3.3v). Interface shall communicate with the meter via PRIA's PACT protocol/DLMS protocol as agreed with SSN.
GPIO	10	Meter services the request and notifies the AMI Module that the meter has an event or alarm requiring service. Meter lowers signal when service request is required and will raise it once the service request has been acknowledge. Meter service request typically occurs within 30 seconds of the occurrence/ detection of an event requiring attention of AMI Module.
GPIO	11	50 Hz zero cross signal when AC mains available in the range of -20% to +15% of nominal voltage.
GPIO	12	NC

An RS-232 interface (RJ-45 socket) is provided under the terminal cover. This port is used for connecting any one of the optional communication modules. Suitable mounting arrangements have been made available on terminal block of the meter for each of these modules.

Table 6: Pin-out details for RJ-45 Socket (for communication when NIC is not used)

Pin No.	Signal at RJ-45	Pin Description
1	Voltage	DC 12 Volts -/+10% to deliver a maximum current of 300mA with voltage and current measured using a True RMS measuring instrument
2	Meter Power Fail /DCD	NA

Pin No.	Signal at RJ-45	Pin Description
3	DTR	Meter transmits / Modem receives It is used to power recycle the modem. Single high pulse of Minimum 1 second duration is provided every Hour on DTR pin (Meter end -RS232)
4	GND	Isolated Ground
5	RxD	Meter receives / Modem transmits. Interface default data rate shall be 9600 bps, 8N1. Interface shall use RS232 signal levels. Interface may multiplex the serial interface between the Modem and the optical port. Interface shall communicate with the meter via PRI's PACT protocol.
6	TxD	Meter receives / Modem transmits. Interface default data rate shall be 9600 bps, 8N1. Interface shall use RS232 signal levels. Interface may multiplex the serial interface between the Modem and the optical port. Interface shall communicate with the meter via PRI's PACT protocol.
7	CTS	NA
8	RTS	NA

5 Energy and Demand Management

Premier 210 provides various time differentiated energy and demand management registers to log various parameters such as energy, demand, etc. It also provides billing support. Details of different registers are tabulated below.

Table 7: Energy and Demand Registers

S. No.	Register Type	Details
1.	Main Registers - Current	Main Energy Registers store values of energy channels irrespective of the time of consumption and indicate cumulative consumption. These registers are available for all supported energy types.
2.	Main Registers - History	Premier 210 maintains one billing (history) register for energy and maximum demand logging. This register is available for all supported energy and demand types.
3.	Rate Register- Current	Rate Registers correspond to a specific period of day as configured under the Seasonal Time of Use (TOU) table. These registers log energy consumption for the current billing cycle for the respective TOU. Six independent Rate Registers are available for a maximum of six supported energy types as configured through M-cubed.
4.	Rate Register- History	These registers are similar to the Current Billing Rate Registers and log the value for previous billing cycles.

In this section you will learn about:

- Energy types supported by Premier 210 under 5.1
- STOU functionality under 5.2
- Time Synchronization functionality under 5.3
- Methods for Billing in Premier 210 under 5.4
- The Maximum Demand functionality under **Error! Reference source not found.**
- The Sampling Window for STOU functionality under **Error! Reference source not found.**
- The Midnight Energy snapshots functionality under **Error! Reference source not found.**
- The Metrological Output functionality under 5.8
- The Remote Firmware Upgrade functionality under 5.9
- The Last Gasp functionality under 5.10
- Day Light Saving Time functionality under 5.11

5.1 Energy Types

Premier 210 supports the following energy channels, out of which up to 16 may be selected for a particular configuration.

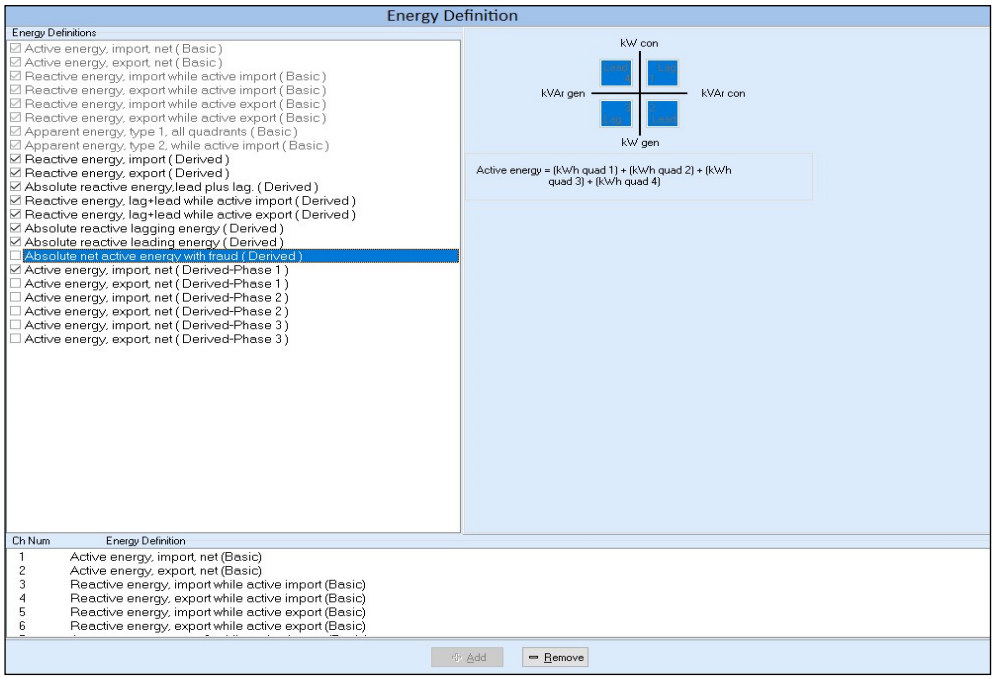
Table 8 Premier 210 Energy Channels

S. No.	Energy Channel	S. No.	Energy Channel
1	Active energy, import, net (Basic)	12	Reactive energy, import while active import (Basic)
2	Active energy, export, net (Basic)	13	Reactive energy, export while active import (Basic)
3	Active energy,import , net (Derived-Phase1)	14	Reactive energy, import while active export (Basic)
4	Active energy,import , net (Derived-Phase2)	15	Reactive energy, export while active export (Basic)
5	Active energy,import , net (Derived-Phase3)	16	Absolute reactive energy, lead plus lag (Derived)
6	Active energy,export , net (Derived-Phase1)	17	Absolute reactive leading energy (Derived)
7	Active energy,export , net (Derived-Phase2)	18	Reactive energy, export (Derived)
8	Active energy,export , net (Derived-Phase3)	19	Reactive energy, import (Derived)
9	Absolute net active energy (Derived)	20	Reactive energy, lag+lead while active export (Derived)
10	Absolute reactive lagging energy (Derived)	21	Reactive energy, lag+lead while active import (Derived)
11	Apparent energy, type 2, while active import (Basic)	22	Apparent energy, type 1, all quadrants (Basic)

For all selected energy channels, two different sets of energy registers are maintained in the meter's memory:

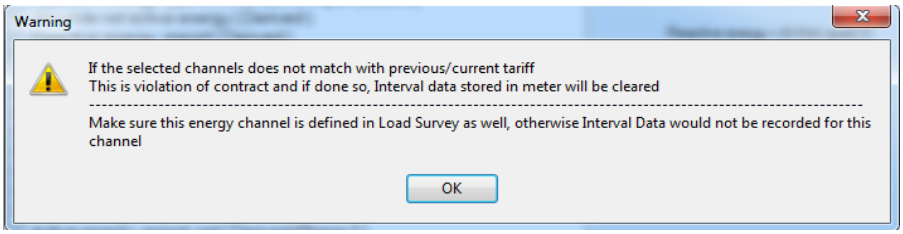
1. Current cumulative values
2. Billing cycle specific values. At billing, a snapshot of energy rate registers is taken and only MD register is reset.

Channels may be specified using the Energy Definition page in the Flextend Configuration Tool from within the M-Cubed software suite.



To add an energy channel, select it and click the **Add** button. The channel is added in the **Energy Definition** selection pane. (Refer figure above). To remove a channel, select it from the **Energy Definition** pane and click the **Remove** button.

Every time a channel is added or removed, a warning message similar to the one below is generated. Please create the energy definition in compliance with the same.



5.2 Seasonal Time of Use (STOU)

Premier 210 supports Seasonal Time of Use and STOU based energy management. The following table describes the STOU components.

Table 9: STOU Components

S. No.	STOU Components	Details
1.	Rate Registers	A maximum of 8 Rate Registers for up to 6 energy channels can be defined for STOU control.

2.	Day Types	A maximum of 16 Day types can be defined. A day-type is defined as a combination of several time intervals spanning 24 hours, each assigned to a specific rate register. A TOU zone can be configured in 1 minute increment settings.
3.	Seasons	A maximum of 13 seasons can be defined. A Season is a non-overlapping period with a start date and an end date. Each weekday of a season is linked with a specific Day-type.
4.	Holidays	A maximum of 40 holidays can be defined. A holiday is assigned a separate day-type which overrides the default day type.

Using these four components, very flexible STOU tariffs can be created. An example is shown in the following table. Here, a day type is defined as a group of multiple time zones spanning 24 hours. Each time zone is assigned with a specific rate register out of the four available rate registers.

A Sample Day-type Configuration

Day Type	Start Time	End Time	Rate Register
1	00:00	07:00	R1
1	07:00	10:30	R2
1	10:30	16:00	R3
1	16:00	22:00	R1
1	22:00	24:00	R1
2	00:00	07:30	R4
2	07:30	10:00	R4
2	10:00	16:00	R5
2	16:00	22:00	R5
2	22:00	24:00	R4

The example above displays only two day types. Up to 16-day types can be defined in a similar manner. Once a day type is defined using time-intervals, seasons can be defined using these day-types; each day-type being previously assigned to days of the week. In other words, each day of a week must first be associated with a specific day type.

Table 10: A Sample Seasons Configuration

Season	Start Date	End Date	Day type Assignment according to Day of Week						
			Mon	*Tue*	*Wed*	*Thu*	*Fri*	*Sat*	*Sun*
1.	01 Jan	31 Mar	1	1	1	1	1	2	2

Season	Start Date	End Date	Day type Assignment according to Day of Week						
			Mon	*Tue*	*Wed*	*Thu*	*Fri*	*Sat*	*Sun*
2.	01 Apr	30 June	3	3	3	3	3	2	2
3.	01 Jul	31 Sep	4	4	4	4	4	5	5
4.	01 Oct	31 Dec	7	7	7	7	7	5	5

The above table illustrates a tariff example where four seasons have been defined. For any holiday that occurs during the four season span, a separate holiday day-type is used. For example, if 25th December is a holiday, then this date is first defined in the holiday definition section and energy consumption is then logged as per the corresponding holiday day-type in the defined rate register.

5.3 Time Synchronization

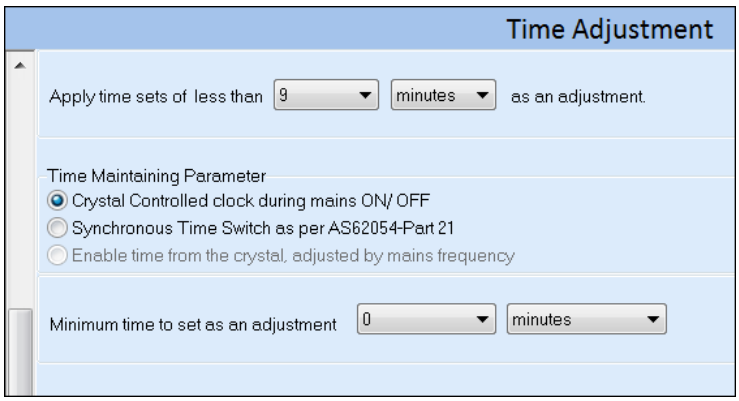
Various meter functions including STOU are time governed. It is therefore essential that the meter clock is accurately synchronized with the real time.

A time adjustment threshold can be defined in Flextend to enable the firmware to distinguish between time adjustment and time set operations.

In response to a forwarded time adjustment request, the meter clock advances after 5 seconds of each minute boundary by upto 50 seconds or less until the time adjustment is complete. For example, if the current meter time is 1 A.M and a forwarded time adjustment request of 60 secs is received, then at 1:00:05 A.M the meter clock will advance by 50 seconds thus setting the clock time to 1:00:55 A.M. The remaining 5 second offset will be applied at 1:01:05 A.M, setting the time to 1:01:10 A.M.

The same principle is followed for a retarding time adjustment, where the meter clock time is decreased gradually by a fixed duration near the minute boundary (in this case the boundary remains constant) until the time adjustment is complete. This eliminates chances of error in activities like demand calculation and load survey logging.

If a time adjustment for more than the defined threshold duration is required, the meter considers it as a time set operation and immediately applies the adjustment.



Any of the following time switches can be used to maintain the meter clock.

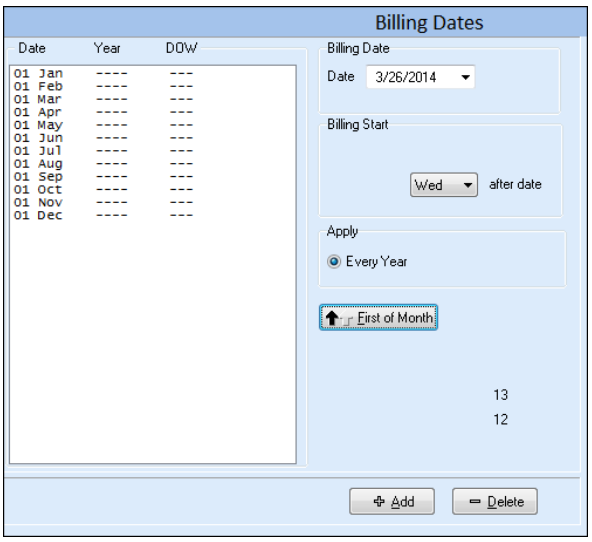
- Quartz crystal controlled time switch (based on Real Time Clock)
- Mains controlled or synchronous time switch (based on line frequency)

You can also configure time maintaining parameters in M-Cubed under **Transactions > Change Operations > Time Maintaining Configuration**. M-cubed first reads the configured parameters from meter and shows them on the GUI. These can be modified as required.

5.4 Methods for Billing

The meter automatically performs a billing action on specified billing dates. The billing action starts a new billing cycle after transferring all billing specific contents to billing or history registers. Billing dates are specified in the configuration. A configuration supports a maximum of 12 or 13 billing dates. Other than this, billing may also occur due to following reasons.

Table 11: Reasons for Billing

S. No.	Billing Reason	Details
1.	Configuration Activation	Automatically at the time of activation of a new or stored configuration.
2.	Software Commands	On receiving authenticated commands from any compatible software like M-Cubed on available communication media.
3.	Engineering Mode	By using push buttons for resetting MD as described under section
For normal billing function, all the required billing dates can be defined through M-Cubed.		

5.5 Maximum Demand

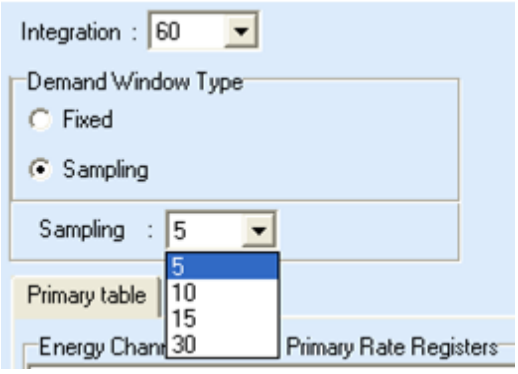
Premier 210 can be configured to record Maximum Demand (MD) for any two of the selected energy channels. Energy demands are monitored periodically at each configured DIP (Demand Integration Period) to record the maximum demand value of each energy type. The DIP can be set to 1, 5, 15, or 30 minutes.

Upon billing, the current values of MD registers are added to the previous cumulative demand registers and the value of MD register is reset. The value of maximum demand can also be reset using an authenticated command from M-Cubed or NMS.

MD values are stored along with the date and start time of the DIP. The maximum value of demand can occur at any time during a DIP, but the meter logs the MD occurrence time as the start time of that respective DIP. For example, if the maximum value of demand occurs at any point of time ranging between 2:30 to 3:00, the meter logs 2:30 as the MD occurrence time.

5.6 Sampling Window for Time of Use

A configuration to record maximum demand on a sampling basis by specifying the sampling demand integration period can be performed in Flextend.



The sampling demand integration period has following list of values:

5, 10, 15 and 30 minutes

For a period of 5 minutes: 1 minute sampling is available.

For a period of 15 minutes: 5 minute sampling window is available.

For a period of 30 minutes: 5, 10 and 15 minutes sampling window is available.

For a period of 60 minutes: 5, 10, 15 and 30 minutes sampling window is available.

NOTE:

DIP value of 1 is allowed for fixed window only.

5.7 Midnight Energy Snapshots

Daily cumulative energy snapshots of configured main energies and TOU parameters are recorded by the meter every midnight. Premier 210 stores upto 35 days of midnight data.

5.8 Metrological Output

Two calibration LEDs are provided on the meter main face plate. They flash in direct proportion to the rate of energy consumption, as determined by the meter pulse weight. Each of these LEDs can be independently assigned to any one of the supported energy channels listed on the **Metrological LED** page in Flextend. The pulse weight is 5000 pulses/unit. The same pulse rate is applied to both the LEDs.

The metrological output from the top LED (Metro LED 1) in the meter can also be directed to the transmission LED of the ANSI port for a predefined time period. The ANSI port is restored to its original function after the default timeout duration of 30 minutes.

The LEDs can also be configured through a transaction in M-cubed under **Transactions > Change Operations > Metro LED Configuration.**

5.9 Remote Firmware Upgrade

Premier 210 supports the functionality of Remote Firmware Upgrade. When the meter receives a remote command, it first checks the authenticity of the data (command) received, and then stores the data in an area different from that of the existing firmware. In this way, two different instances of firmware reside in the meter memory – one being executed and the other waiting to be invoked.

It is possible to enable the new firmware remotely. A remote firmware upgrade does not affect the metrology of the meter nor does it cause any loss of interval data, event logs or stored configurations during the firmware change. Also, it does not make any difference in recording the energy consumption on all channels during the process. If the firmware upgrade is unsuccessful the meter will revert back to the previous firmware.

The firmware downloaded remotely in the meter will be immediately activated.

5.10 Last Gasp

When the meter detects a power outage, it sends a message to the NIC, which then starts an orderly shutdown. This includes transmitting an outage notification to the Access Point (AP) which, in turn, transmits it further to the back office.

NOTE:

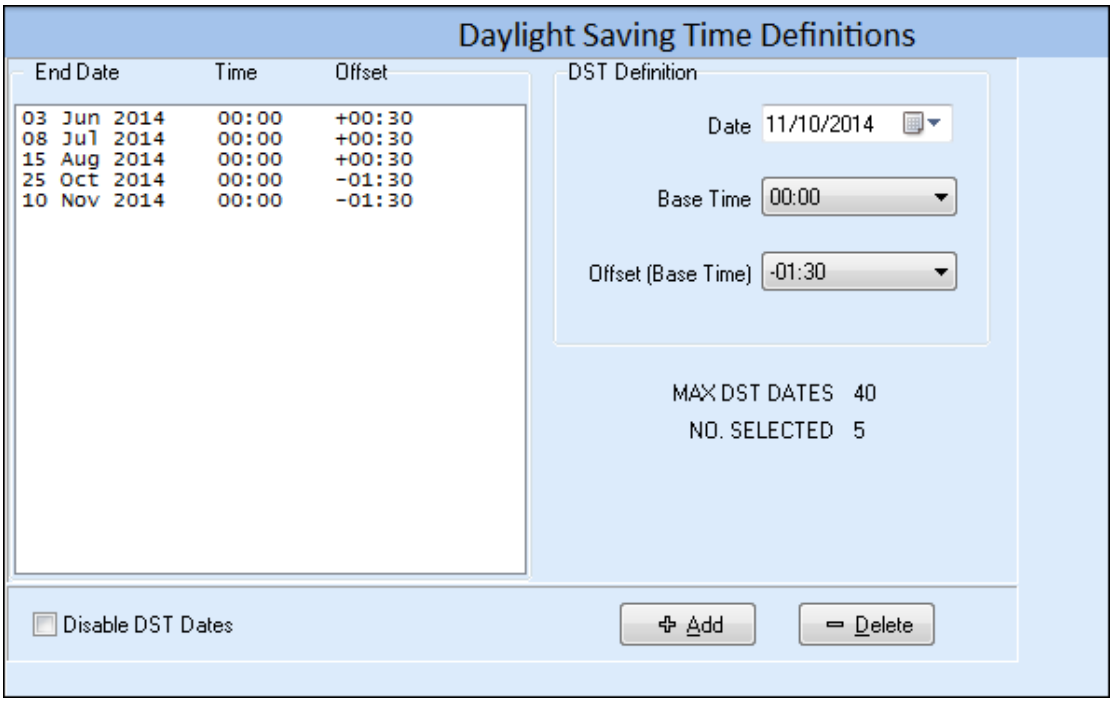
This feature is applicable only for AMI metering.

5.11 Daylight Saving Time (DST)

To support Daylight Savings Time, the meter distinguishes between Base Time and Seasonal Time. The seasonal time is maintained according to DST as defined in the tariff configuration in Flextend. DST is defined using the base date, end time of a seasonal time and a DST offset. Premier 210 can be configured to display both base time and seasonal time.

While the base time is always maintained by the metering firmware, a seasonal time is maintained according to the DST definition provided in configuration. For example a maximum of 40 DSTs can be set to move the meter clock forward or backward by half-an hour or more to adjust the meter time where applicable.

Premier 210 automatically distinguishes between base time and seasonal time. All STOU controlled activities such as rate register energy consumption logging, MD logging, and load control are performed according to the seasonal time. However, interval data, events are always logged according to the base time.


Daylight Saving Time Definitions
End Date
Time
Offset
03 Jun 2014 00:00 +00:30
08 Jul 2014 00:00 +00:30
15 Aug 2014 00:00 +00:30
25 Oct 2014 00:00 -01:30
10 Nov 2014 00:00 -01:30
DST Definition
Date
11/10/2014
Base Time
00:00
Offset (Base Time)
-01:30
MAX DST DATES 40
NO. SELECTED 5
Disable DST Dates
Add
Delete

6 Condition Monitoring and Event Logging

The monitoring functions in Premier 210 allow displaying, logging and communicating different conditions of the meter to the user. These can be categorised as follows:

Table 12: Condition Monitoring and Event Logging

S. No.	Function	Details
1.	Instantaneous Values	Instantaneous values for different electrical parameters such as voltage, current, power factor, power, frequency, harmonic distortions etc. are displayed on the meter and are also communicated to the compatible software at the back office.
2.	Events	Events are logged for either one or a combination of multiple predefined operating or electrical conditions. Sprint detects such conditions and stores them with occurrence and restoration time. In addition to a standard event logger, Premier 210 supports separate loggers for power quality, power failure, fraud detection and switching events. The logged events in the meter can be communicated to the back office through compatible software. It is also possible to store average values and extreme drifted values of other electrical parameters at the time of event occurrence. An alarm facility can be configured for required events in M-Cubed.
3.	Flags	A flag is a Boolean value which indicates either an occurrence or a restoration of a special condition / event / action / status. Sprint displays, logs and communicates these to compatible software.
4.	Load Survey	Load Survey is the consumption value logged for consecutive intervals, also identified as 'Survey Integration Period' (SIP). Additionally, a flag can also be associated and logged with a survey integration period.

In this section you will learn about:

- Instantaneous value monitoring functionality in Premier 210 under **Error! Reference source not found.**
- The Event monitoring functionality in Premier 210 under **Error! Reference source not found.**
- Load Survey/Interval Data functionality in Premier 210 under **Error! Reference source not found.**
- Meter Security in Premier 210 under **Error! Reference source not found.**
- The QOS functionality in Premier 210 under **Error! Reference source not found.**

6.1 Instantaneous Values

Premier 210 supports logging of a complete set of instantaneous electrical parameters such as power, voltage, current, frequency, power factor etc. These parameters are preconfigured for being displayed on the meter display and are also used for other monitoring functions of meter such as event snapshot and QOS Event feature.

In addition to the meter display, instantaneous values can also be read using software like M-Cubed, which show the values of the parameters with a time stamp i.e. with the time when such parameters were logged.

6.2 Events

Apart from logging electrical energy and billing data, the meter can also be configured to record a predefined operation or electrical condition identified as an event. These events are used to detect different conditions such as attempted tampers, QOS monitoring and electrical condition monitoring. Premier 210 detects and logs such conditions with special logic built into the metering software. The meter has five separate loggers for events related to standard conditions, power quality, power failure, fraud detection and contactor (switch) operations.

S. No.	Event Log	Details	No of Supported Events
1.	Standard	Logs events related to general meter conditions such as Low Battery, Wrong Phase Sequence.	500
2.	Power Quality	Logs events related to energy distribution network such as Over Voltage, Current Reversal, or Under Voltage Restoration.	450
3.	Power Failure	Logs events related to outages such as L1 Voltage Missing/failure, L3 Voltage Missing/failure restoration.	400
4.	Fraud Detection	Logs events related to tamper and fraud conditions such as Front Cover removed, Module Cover replaced.	500
5.	Switch	Logs events related to use of the supply and load contactors such as Meter Disconnection, Meter connection.	500
6.	Current	Logs events related to changes in flow of current such as over or reverse current.	450
7.	Duration	Logs events related time	400

The meter logs time stamped values for both occurrence and restoration of an event. For occurrence, an event is logged if it persists for a duration greater than a predefined time identified as ’persistence time-occurrence’. Similarly, when the conditions return to normal and stay normal, the meter logs an event after a predefined period identified as ’persistence time-restoration’.

If an event is configured for alarm, the meter signals this to NIC module which reads the event and communicates it to the NMS.

6.3 Load Survey / Interval Data

Load profile data comprises load consumption and instantaneous parameter values recorded by the meter at consecutive trading intervals, and can be set-up through a configuration in Flextend from within the M-Cubed software suite. This interval is identified as ‘Survey Integration Period,’ or SIP. Additionally, it is also possible to log a flag corresponding to a SIP for an occurrence of an event. The significance of the different load survey values recorded w.r.t. SIP is shown below:

Table 13: Different Load Survey Values Recorded with Respect to SIP

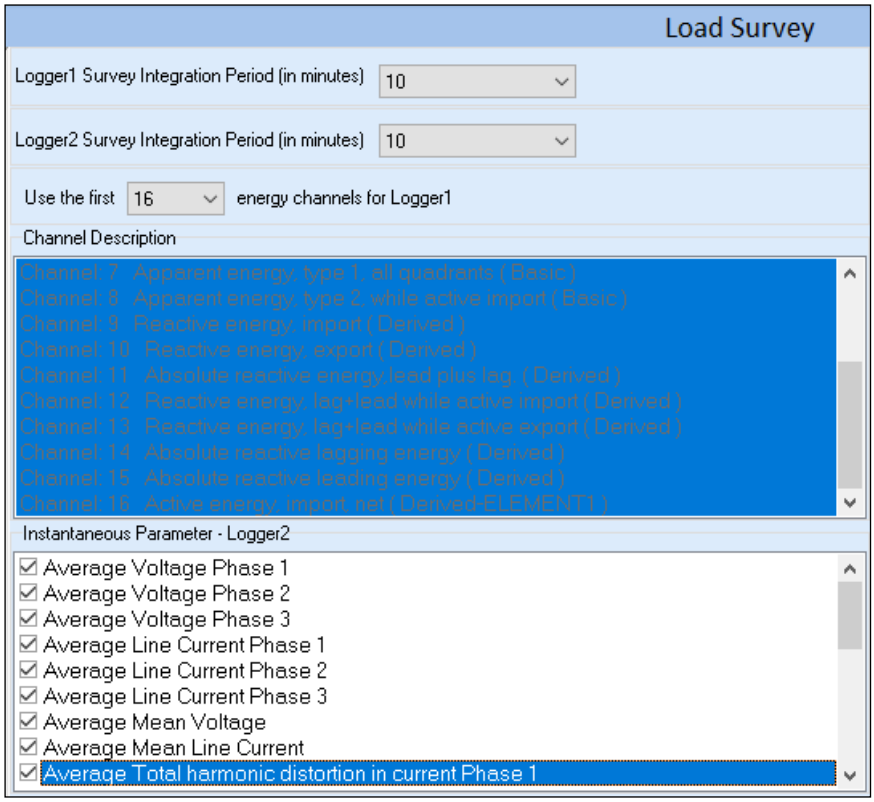
Sr. No	Bit	Bit Name	Bit Status	Indication of Bit
1.	0	Absent reading	CLR	Normal data recording
			SET	Data is missing due to power failure
2.	1	Incomplete Interval	CLR	Complete recording interval
			SET	Interval becomes short due to time synchronization
3.	2	Power failed	CLR	Normal data recording
			SET	Indicates power was lost / restored during the interval.
4.	3	Time set	CLR	Normal
			SET	Indicates time set was done in this SIP
5.	4	Not used	CLR	
			SET	
6.	5	Reverse Run	CLR	Current for all phases is forwarded
			SET	Current is reversed for individual phase
7.	6	Phase Missing	CLR	All phase voltages are present
			SET	Voltage missing for a phase
8.	7	Data change	CLR	Clear in all other cases
			SET	In case of data change due to Configuration change, MD reset by push button or MD reset by communication and register reset by communication

Premier 210 has two load profile loggers which work independently of each other.

- **The primary logger:** Can be configured to capture load profile data for upto 16 energy channels at programmable intervals ranging from 5 to 60 minutes. It logs data for 3600 parameters days with a 30 minute SIP.
- **The secondary logger:** Can be configured to capture load profile data for upto 16 out of the 24 instantaneous parameters at programmable intervals ranging from 1 to 60 minutes. It logs data for 4320 parameters days with a 30 minute SIP.
- Refer **Error! Reference source not found.** for the list of instantaneous parameters supported by Premier 210.

The number of days of load profile data that the meter can record depends on the SIP as well as the number of energies and instantaneous parameters configured in the meter. For example, 2-channel, 15-minute data can be retained for two months and 2- channel, 1-hour data can be retained for 12 months, when using the recommended memory configuration. Meters support viewing load profile data on the

meter display. The Survey Integration Period for the primary logger 1 is: 5, 10, 15, 30 and 60 minutes. The Survey Integration Period for the secondary logger 2 is: 1, 5, 10, 15, 30 and 60 minutes.



NOTES:

- Both the primary and secondary loggers will record time stamped load profile values at the end of each SIP. It is possible to read load survey data from a defined SIP on a particular date to a desired SIP on some other date. In this case the time stamped values will be shown only for the first read SIP and the first SIP of each day for the specified date range.
- LS day cross over will happen in accordance with pre-configured time duration.
- All previous Load Survey data will be lost when a new LS configuration is loaded, so be sure to read the data before commencing a new configuration.
- If the SIP for only one of the two loggers is changed, the previous data for impacted logger will be erased.

6.4 Meter Security

The Premier 210 meter is provided with various security options to avoid physical tampering, data corruption and unauthenticated access. Some of the security options are tabulated below:

Table 14: Premier 210 Meter Security Features

S.No.	Functions	Details
1.	Sealing Feature	To prevent physical tampering, an adequate sealing arrangement is provided on the meter cover and terminal cover. An additional sealing arrangement is provided on the **Config** button that prevents unauthorised access to Engineering mode.
2.	Cover Open Detection	The meter detects opening of the front and terminal cover even when meter is switched off and maintains a log of cover open events in its memory.

You can read and modify authorisation time for front cover open in M-Cubed under **Reset > Authorize Front Cover Open.** This enables distinction between authorized and un-authorized terminal cover open activity.

The data in the Premier 210 meter is protected at various security levels as explained below:

Table 15: Premier 210 Meter Security Levels

Client Server Association Level	Security Level	Type	Operations	Usage
1.	Level 0 (No security)	Public Client – General Meter Information (Read Only Access)	Read general meter information such as, Meter serial no, time and device ID.	Access to meter information such as Meter serial no, time and ID.
2.	Level 1 (low level)	Meter Reader- Meter Data Reading (Read Only Access with password)	Meter Reading	In addition to level 0 activities, read only access to Instantaneous, LS, Billing, Daily Energy, Events, and Transaction data etc.
3.	Level 2 (High level)	Utility Client- Meter Data Reading and Transactions (Administrative Tasks)	Meter reading, meter settings such as time / date, MD reset, changing association level 1 password	In addition to level 1 activities, new meter time / date can be set and MD registers can be reset. New passwords for lower levels (level 1) can be set.

4.	Level 3 (High level)	Management Client- Meter Data Reading and Transactions (Administrative Tasks)	Meter reading, meter settings such as time / date, MD reset, changing association level 1 password. Has access to Remote port and BCS	In addition to level 1 and 2 activities, new meter time / date can be set and MD registers can be reset. New passwords for lower levels (level 1) can be set.
5.	Level 4 (High level)	Remote Port Reader Client- Meter Data Reading, Configuration, and Transactions	Meter reading, meter settings such as time / date, MD reset, changing association level 1, 2 passwords, resetting registers, downloading tariff files etc.	In addition to level 2 and 3 activities, tariffs can be downloaded. New passwords for lower levels (level 1, 2) can be set.
6.	Level 5 (High level)	Remote port Utility Client –All meter operations.	Meter reading, meter settings such as time / date, MD reset, changing passwords for association levels 1, 2, 3, and 4, calibrating meter constants, resetting registers, downloading/replacing tariff files etc.	All transactions can be performed with the meter at this level of authorization.

7 Pulse Output Functions (Optional)

Premier 210 provides one pulse output with voltage ratings of up to 27V DC at a current rating of 27mA.

The outputs in the meter can be configured to indicate either an event or state.

- State: The State type output refers to a definite state i.e. ‘ON’ or ‘OFF’ at the port
- Event: The Event type output refers a pulse of a short interval and returns back to its normal state.

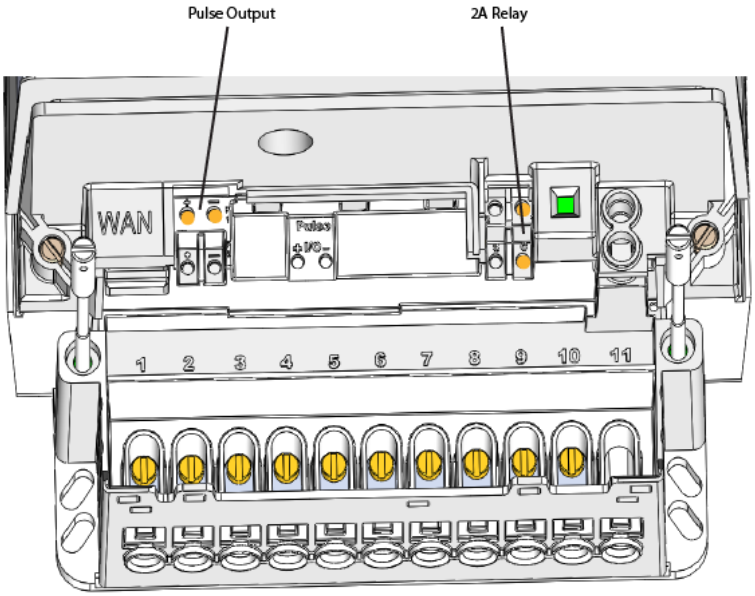


Figure 5: Terminal block indicating the pulse output terminals

One output terminal as shown in the figure provided above are used as potential free outputs.

Table 16: Pulse Output Combination

Option	P1	P2
Output - Dry	27 V DC	230 V AC

The pulse outputs can be individually configured to provide output under the control of metering software.

Following is the list of different output functions and their significance. These functions are supported according to the specific hardware build and meter firmware.

Table 17: List of Different Output Functions and their Significance

S. No.	Function	O/P Type	Details/ Remarks
1.	Energy Pulse	50 ms pulse	Proportional to energy recorded for a selected energy channel. Number of pulses corresponds to the configured pulse-weight.
2.	DIP Crossover	0.5 s to 10 s pulse	One pulse at Demand Integration Period (DIP) crossover

S. No.	Function	O/P Type	Details/ Remarks
3.	Switching of Rate Register/Demand Register	0.5 s pulse	One pulse at rate register crossover
4.	STOU Output	State – On	Output can be configured to maintain ON state in accordance with STOU switching arrangement
5.	Energy Register / Demand Registers	State – On	Output can be configured to maintain ON state, if any of the selected rate registers is active
6.	Time sync pulse	Pulse	Output can be configured to provide a pulse at some predefined interval for time synchronizing
7.	Billing Action	Pulse	Output can be configured to provide a pulse at the time of billing action
8.	Configurable time period	0.5 to 10 s pulse	Output can be configured to provide a pulse at the end of a configurable time period

7.1 Pulse Output Configuration according to other functions

Output pins under other functions can be configured under the **I/O Configuration** page in Flextend. Based on the function selected, function specific options are displayed. It is not possible to apply more than one function on the same pin.

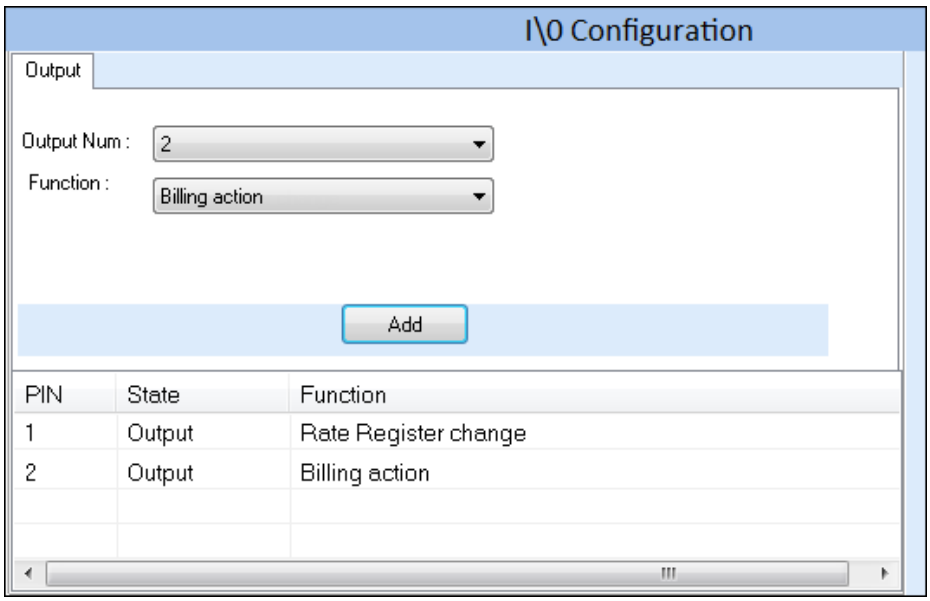


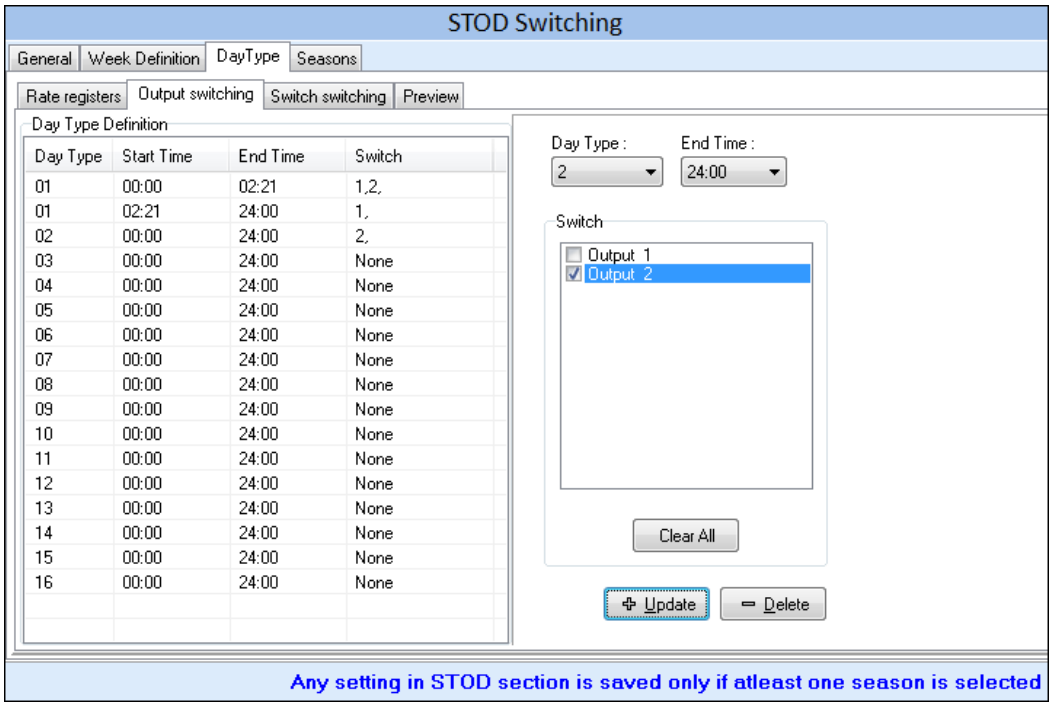
Table 18: Pulse Multiplier and Corresponding Energy Usage for Pulse Configured

Multiplier	Energy per pulse
1/10th	1000 Wh
1/5th	500 Wh

1/4th	400 Wh
1/2nd	200 Wh
1	100 Wh
2	50 Wh
4	25 Wh
5	20 Wh
10	10 Wh

7.2 Pulse Output Configuration under the STOU Control

Using M-Cubed, output pins can be configured under the **Output Switching** tab for Day Type Definition under the **STOD Switching** page.



8 Local and Remote Communication

Premier 210 supports configurable baud rate for local and remote communication. The baud rate can be configured in Flextend or changed in the field through a transaction via M-Cubed.

The default and configurable baud rates by communication port types are tabulated below:

Table 19: Communication Interface Details

Channel	Default Baud Rate (bps)	Configurable Baud Rate (bps)	Remarks
ANSI Optical Port	9600 bps	9600, 19200, or 38400, 57600	Direct Communication, HDLC (DLMS)
Remote	9600 bps	9600, 19200, or 38400	Direct Communication, HDLC (DLMS)

9 Engineering Menu

The Engineering Menu allows performing different operations on Premier 210 such as performing an LED or Contactor Test, altering the active configuration in the meter such as resetting the MD, configuring phase(s) or testing the accuracy of the meter in test mode.

Please note the following in connection with the Engineering menu:

- To evoke the Engineering mode, press the **Config** (left) push button for five seconds. The same button will allow navigation through all the menu options supported in Engineering Mode.

NOTE:

To help prevent unauthorised use of the meter using the push buttons, the meter comes with a default terminal cover interlock. To access the Engineering menu for the first time, open the main terminal cover and replace it back. An event for terminal cover open will be logged by the meter.

- Push the **Scroll** (right) push button to access the required engineering menu option.
- Push the **Config** button to navigate/advance through the sub options of an engineering menu option.

In this section you will learn to:

- Configure an Energy Channel for Metro LED in the Engineering Mode under 9.1
- Configure the meter for phase wise metering in the Engineering Mode under 9.2
- Reset the MD in the Engineering Mode under 9.3
- Perform a Contactor Test in the Engineering Mode under 9.4
- Manually disconnect/re-connect meter supply under **Error! Reference source not found.**
- Check the meter accuracy in Test Mode under 9.5
- Select and activate a stored tariff under 9.6

A complete list of sample display screens are shown under section **Error! Reference source not found.**.

9.1 Energy Channel Configuration for Metro LED

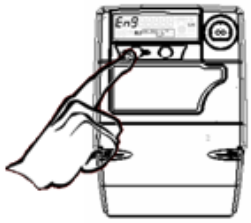
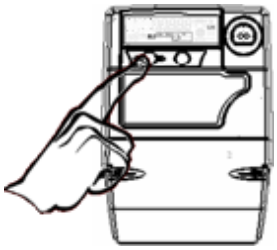
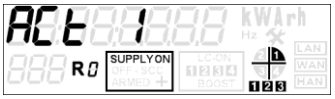
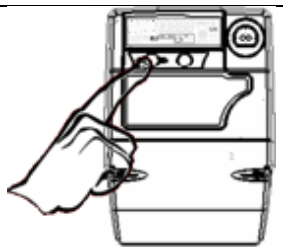
Two LEDs are provided on the meter with default settings - one for Active and the other for Reactive energy. Once configured, they flash in direct proportion to the rate of selected energy consumption; as determined by the meter pulse.

Other than the default, these LEDS can be temporarily configured to flash on any one of the 6 supported energy channels through the Engineering mode.

NOTE:

After a Power cycle, the LED flashing resorts to default mode.

The procedure to configure an energy channel for Metro LED through Engineering mode is as follows:

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode.		
Step 2: Navigate to the LED test option.		
Step 3: LED Test option - press the **Scroll** button. This action will invoke a list of the energy parameters configured in the meter using Flextend. An example parameter is shown here.		
Step 4: To navigate through the list press the **Config** button. Navigate to required parameter for taking the LED test. This example depicts the test for the Active Energy Export parameter.		
Step 5: Press the **Scroll** button to start the test. The **Accept** display will appear.		

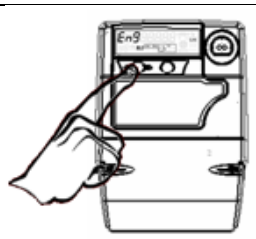
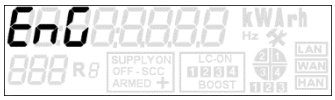
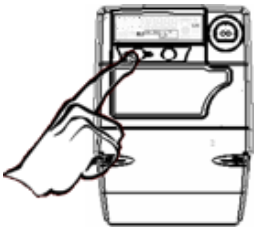
Step 6: Press the **Scroll** button to accept the configuration. The configured LED will start flashing for that parameter depending on the meter load condition.		

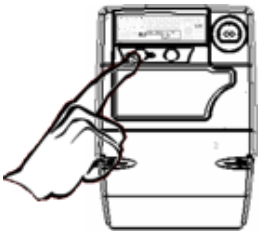
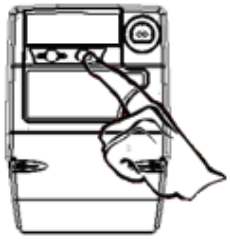
NOTE:

On completion of the configuration, revert to the normal mode by pressing the **Scroll** button.

9.2 Phase Configuration

Premier 210 can be configured to function on supply phase A, B and C or a combination of these. The procedure for configuration is shown below:

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode.		
Step 2: Navigate to the Phase Configuration option.		
Step 3: Phase Configuration option - press the **Scroll** button. This action will invoke a list of supply phase combinations that can be metered by the device. An example combination is shown here.		

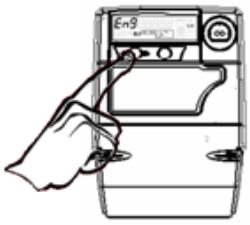
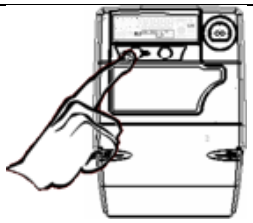
Step 4: To navigate through the list press the **Config** button. This example depicts meter configuration for two phase combination on B-C phases.		
Step 5: Press the **Scroll** button to accept the configuration. The Accept display appears.		

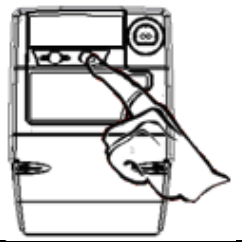
NOTE:

On completion of the configuration, revert to the normal mode by pressing the **Scroll** button.

9.3 MD Reset

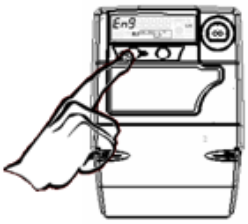
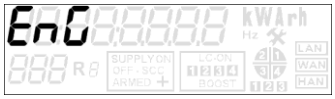
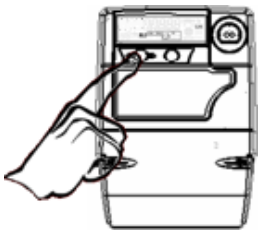
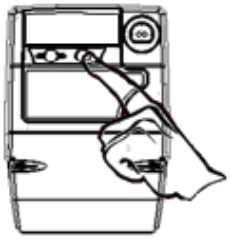
At the time of a billing action, the current values of Energy, Rate and Demand registers are moved to a history register and the value of the MD register is reset. The procedure to reset the MD is described below:

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode.		
Step 2: Navigate to the MD Reset option.		
Step 3: MD Reset option - press the **Scroll** button. This action will invoke the bill display.		

Step 4: Press the **Scroll** button again to reset the MD. A confirmation screen is displayed.		

9.4 Contactor Test

This test validates if the contactor state (on/off) changes in response to the issued command. The procedure to perform the Contactor test is as follows:

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode.		
Step 2: Press the **Config** button again and navigate to the Contactor Test option.		
Step 3: Contactor Test option - press the **Scroll** button to enter the option menu.		
Step 4: Press the **Scroll** button to complete the test. The desired switch is now connected.		

Notes:

The Contactor test is available for the following switch types:

- Switch 2 or Relay 1 (Build Specific)

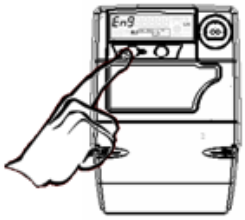
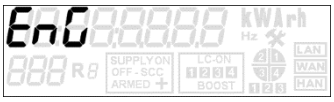
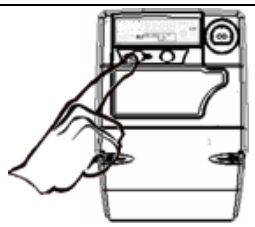
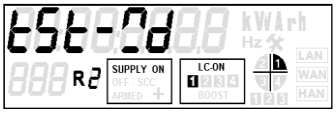
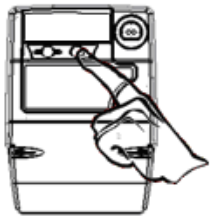
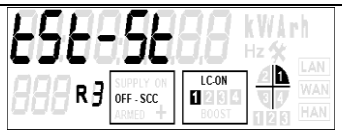
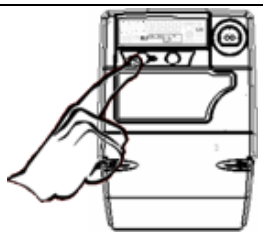
On completion of the test, revert to the normal mode by pressing the **Scroll** button.

9.5 Test Mode

This option allows checking the accuracy of the metering device against a known load condition. The test can be performed verifying the direction, current, voltage or energy values logged by the meter against a standard input.

Energy increment in the Main energy register is not stopped during test mode and the incremental energy value for Active Import Total and Active Export Total is displayed in high-resolution.

The procedure to perform the test is shown below:

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode.		
Step 2: Navigate to the Test Mode option.		
Step 3: Test Mode option - press the **Scroll** button. The Start Test display will appear.		
Step 4: Press the **Scroll** button again. The Accept display appears. This indicates that the test has started.		
Step 5: To end the test follow steps 1 to 3. Next click the **Config** button. The End Test display will appear.		

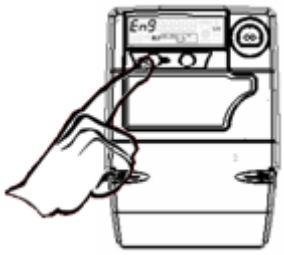
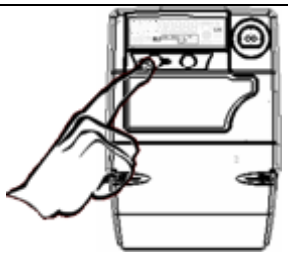
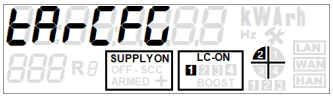
Step 6: To end the test press the **Scroll** button. The **Accept** display appears. This indicates that the test has ended.		

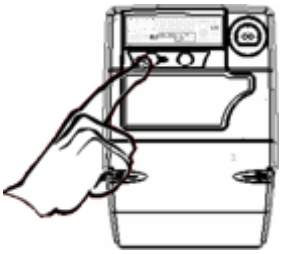
NOTE:

During or on completion of the test navigate to the main energy display for Active Import Total or Active Export Total to view the changing values as a result of incrementing energy registers. An example display is shown for Active Import Total.

9.6 Tariff Activation

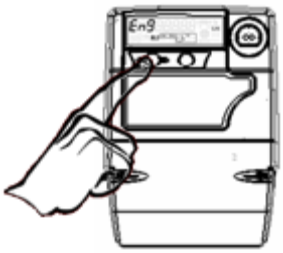
This option allows selection of a tariff from the meter's memory and further activating it.

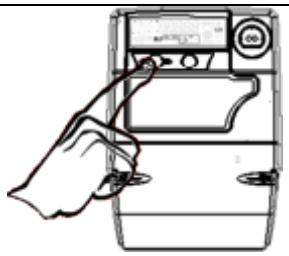
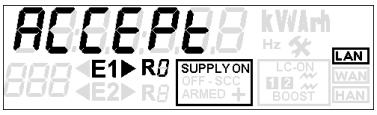
Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode.		
Step 2: Navigate to the Tariff Configuration option.		

Step 3: Tariff Configuration option - press the **Scroll** button. This action will invoke a list of stored tariffs available in the meter. An example tariff is shown here.		
Step 4: To navigate through the list press the **Config** button and select the desired tariff.		
Step 5: Press the **Scroll** button to accept the tariff configuration. The **Accept** display appears confirming acceptance.		

9.7 Antenna switching

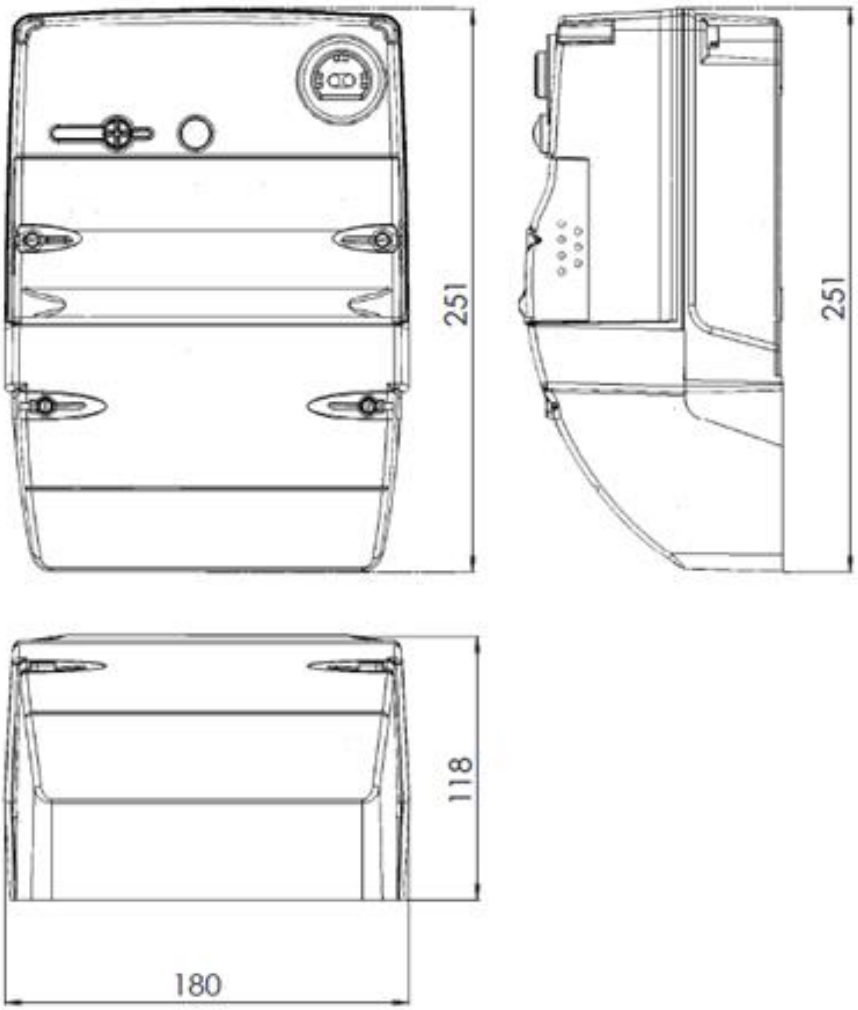
This option allows switching the NIC antenna from to external and vice-versa.

Step 1: Press the **Config** button for five seconds to evoke the Engineering Mode. The same button will enable navigation through all the options supported in Engineering Mode. Navigate to the NIC CF option.		

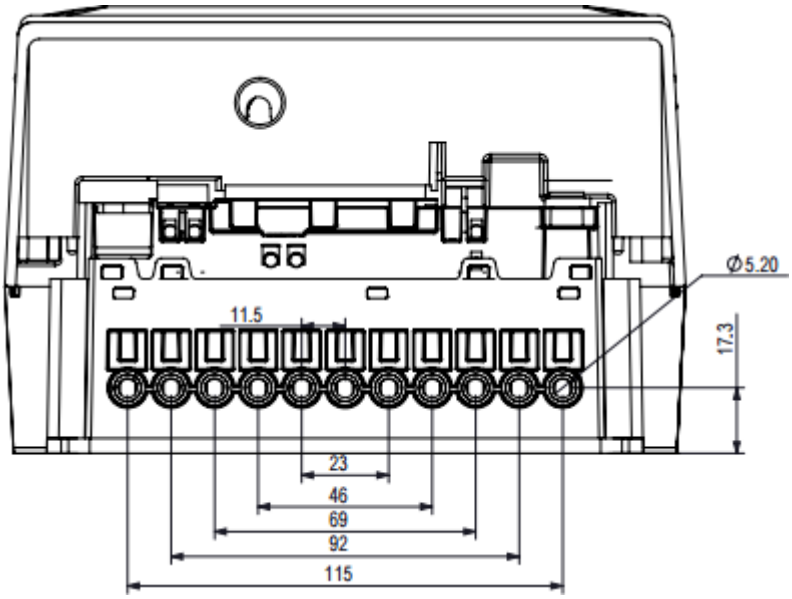
Step 2: Press the **Scroll** button to select the NIC- CF option.		
Step 3: Press the **Config** button to navigate between these two screens.		
Step 4: Press the **Scroll** button to select the desired antenna type. The **Accept** display appears confirming acceptance.		

10 Premier 210 Dimensions and Mounting Details

Premier 210 supports various mounting options which include arrangements to substitute any currently installed meters too. The figure below displays the rear view fixing dimensions in mm.

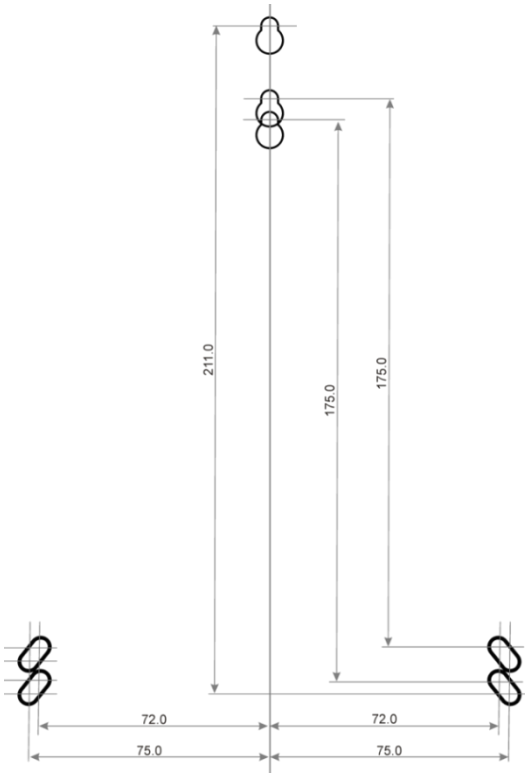


Terminal connections



Dimensions and Mounting Details

I



Mounting Arrangements

Appendix 1. LCD Displays

Given below is a list of displays supported by Premier 210. Some of the displays are variant specific and may not be available in all variants.

1	CHANNEL REGISTERS		2	CHANNEL REGISTERS-HISTORY	
STANDARD ENERGY REGISTERS	123456 kWh 001 R2	active import energy	HISTORY (MOST RECENT) ENERGY REGISTERS	98765 kWh 028 R2	active import energy-History 1
	123456 kWh 002 R2	active export energy		98765 kWh 029 R2	active export energy-History 1
	123456 kVArh 003 R2	reactive import energy (Q1+Q2)		98765 kVArh 030 R2	reactive import energy (Q1+Q2) - History 1
	123456 kVArh 004 R2	reactive export energy (Q3+Q4)		98765 kVArh 031 R2	reactive export energy (Q3+Q4) - History 1
	123456 kVAh 007 R2	Apparent (All Quadrants)		98765 kVAh 034 R2	Apparent (All Quadrants)History 1
	123456 kVAh 008 R2	Apparent (while Active Import		98765 kVAh 035 R2	Apparent (while Active Import History 1
	123456 kWh 012 R2	Active energy (kWh) Import – Phase1		98765 kWh 039 R2	Active energy (kWh) Import – Phase1History 1
	123456 kWh 013 R2	Active energy (kWh) Import – Phase2		98765 kWh 040 R2	Active energy (kWh) Import – Phase2History 1
	123456 kWh 014 R2	Active energy (kWh) Import – Phase3		98765 kWh 041 R2	Active energy (kWh) Import – Phase3History 1
	123456 kWh 015 R2	Active energy (kWh) Export – phase1		98765 kWh 042 R2	Active energy (kWh) Export – phase1History 1
	123456 kWh 016 R2	Active energy (kWh) Export – phase2		98765 kWh 043 R2	Active energy (kWh) Export – phase2History 1
	123456 kWh 017 R2	Active energy (kWh) Export – phase3		98765 kWh 044 R2	Active energy (kWh) Export – phase3History 1

	123456 kWh 018 R2	Active energy(kWh) imp+Exp.(Forwarded) - 3Phase		98765 kWh 045 R2	Active energy(kWh)imp+ Exp.(Forwarded) - 3PhaseHistory 1
	356478 kVArh 046 R2	Reactive energy Import while Active Import (Q1)		256478 kVArh 061 R2	Reactive Energy Import while Active Import (Q1) -History 1
	356478 kVArh 047 R2	Reactive Energy Export while Active Import (Q4)		256478 kVArh 062 R2	Reactive Energy Export while Active Import (Q4) - History 1
	356478 kVArh 048 R2	Reactive Energy Import while Active Export (Q2)		256478 kVArh 063 R2	Reactive Energy Import while Active Export (Q2) - History 1
	356478 kVArh 049 R2	Reactive Energy Export while Active Export (Q3)		256478 kVArh 064 R2	Reactive Energy Export while Active Export (Q3) - History 1
3	TARIFF RATE REGISTERS		4	TARIFF RATE REGISTERS-HISTORY	
STANDARD TARIFF RATE REGISTERS	356478 kWh 040 R2	active import energy	HISTORY (MOST RECENT) TARIFF RATE REGISTERS	256478 kWh 055 R2	active import energy-History 1
	356478 kWh 041 R2	active export energy		256478 kWh 056 R2	active export energy-History 1
	356478 kVArh 042 R2	reactive import energy (Q1+Q2)		256478 kWh 057 R2	reactive import energy (Q1+Q2) - History 1
	356478 kVArh 043 R2	reactive import energy (Q3+Q4)		256478 kWh 058 R2	reactive import energy (Q3+Q4) - History 1
	356478 kVAh 050 R2	Apparent (All Quadrants)		256478 kVAh 065 R2	Apparent (All Quadrants) - History 1
	356478 kVAh 051 R2	Apparent (while Active Import)		256478 kVAh 066 R2	Apparent (while Active Import) - History 1
	356478 kVAh 054 R2	Current/active rate register			
5	DIAL TEST REGISTERS (HIGH RESOLUTION ENERGY REGISTERS)				
	12245879 Wh 031 R2	active import energy		12245879 VArh 039 R2	Reactive Energy Import while Active Export (Q2)
	12245879 Wh 032 R2	active export energy		12245879 VArh 040 R2	Reactive Energy Export while Active Export (Q3)
	12245879 VArh 033 R2	reactive import energy (Q1+Q2)		12245879 VAh 041 R2	Apparent (All Quadrants)

	12245879 VArh 034 R2	reactive import energy (Q3+Q4)		12245879 VAh 042 R2	Apparent (while Active Import)
	12245879 VArh 037 R2	Reactive Energy Import while Active Import (Q1)		12245879 VArh 038 R2	Reactive Energy Export while Active Import (Q4)
6	MAXIMUM DEMAND (MD)				
ACTIVE ENERGY FOR MDREGISTER	13.786 kW 070 R2	UMD register for channel 1	REACTIVE ENERGY FOR MDREGISTER	19.786 kVAr 072 R2	UMD register for channel 1
	19.786 kW 071 R2	UMD register for channel 2		19.786 kVAr 073 R2	UMD register for channel 2
APPARENT ENERGYFOR MD REGISTER	22.790 kVA 074 R2	UMD register for channel 1			
7	RISING DEMAND				
RISING DEMAND	28: 02.786 kW 070 R2	rising demand (active import energy)	RISING	28: 02.750 kW 071 R2	rising demand (active export energy)
	28: 02.750 kVAr 072 R2	rising demand (reactive energy)		28: 03.750 kVA 073 R2	rising demand (Apparent energy)
8	MAXIMUM DEMAND TIME AND DATE				
	02:30:00 t kVA 080 R2	MD time (kVA resembles energy channel i.e whether active, reactive or apparent)		10: 12: 11 d kVA 081 R2	MD date
9	MAXIMUM DEMAND RESET DATE, RESET TIME AND RESET COUNT				
MD RESET DATE	10: 11: 11 d kWArh 083 R2	date of billing event	MD RESET	10:30:00 t kWArh 085 R2	time of billing event
	-------- kWArh 083 R2	billing event has not occurred		-------- kWArh 084 R2	billing event has not occurred
	bi LL: 4 kWArh 085 R2	MD reset count			
10	TIME AND DATE DISPLAY				
	10: 15:54 t kWArh 086 R2	standard time (24 hours clock)		10: 11: 11 d kWArh 087 R2	standard date
	-------- kWArh 094 R2	bad time (clock failure)		-------- kWArh 094 R2	bad date (clock failure)
11	CURRENT DISPLAYS				
	1A 76.45 A kWArh 015 R2	active current (phase 1)		100.00 A kWArh 011 R2	over current threshold

	2A-076.45 012 R2	active current (phase 2)		3A 53.52 016 R2	active current (phase 3)
	1r 2.67 013 R2	reactive current (phase 1)		2r 12.67 017 R2	reactive current (phase 2)
	3r 2.67 017 R2	reactive current (phase 3)		1L 97.67 019 R2	line current (phase 1)
	2L 57.70 016 R2	Line current (phase 2)		3L 57.80 019 R2	line current (phase 3)
	3P 100.0 019 R2	maximum current (3 phase)		P1 100.0 030 R2	maximum current (phase 1)
	P2 100.0 027 R2	maximum current (phase 2)		P3 100.0 031 R2	maximum current (phase 3)
	3P 5.0 021 R2	minimum current (3 phase)		P1 1.0 022 R2	minimum current (phase 1)
	P2 2.0 029 R2	minimum current (phase 2)		P2 7.0 033 R2	minimum current (phase 3)
12	POWER FACTOR DISPLAYS				
	3P 1.000 033 R2	maximum power factor (3 phase)		P1 1.000 043 R2	maximum power factor (phase 1)
	P2 1.000 039 R2	maximum power factor (phase 2)		P3 1.000 044 R2	maximum power factor (phase 3)
	3P 0.525 040 R2	minimum power factor (3 phase)		P1 0.035 045 R2	minimum power factor (phase 1)
	P2 0.520 042 R2	minimum power factor (phase 2)		P3 0.035 047 R2	minimum power factor (phase 3)
13	POWER DISPLAYS				
	72.000 kW 055 R2	active power (instantaneous)		12.528 kVAr 056 R2	reactive power (instantaneous)
	76.545 kVA 057 R2	apparent power (instantaneous)		P1 10.300 kW 058 R2	Phase1 active power (instantaneous)
	P2 10.300 kW 059 R2	Phase2 active power (instantaneous)		P3 10.300 kW 060 R2	Phase 3 Active Power(instantaneous)
	P1 10.300 kVAr 061 R2	Phase1 reactive power (instantaneous)		P2 10.300 kVAr 062 R2	Phase2 reactive power (instantaneous)
	P3 10.300 kVAr 063 R2	Phase3 reactive power (instantaneous)			
14	GENERAL DISPLAYS				

	PAGE 01 888 R8	page		LEd 6000 kWh 888 R8	Metrological LED
	240.0 V 012 R8	primary voltage		100.00 A 013 R8	primary current
	100.00 A 014 R8	current limit value		180.0 V 015 R8	voltage limit value
	FPG 1.001 015 R8	Clem name		GroUP000 015 R8	tariff name
	10:38 015 R4	remaining time adjustment		-01:56 016 R4	remaining time adjustment (reverse time)
	888 R8	blank		USn00008 018 R4	serial number (Low)
	EnGY 017 R4	serial number (High)		888 R8	Display ID
15	METER STATUS				
	St:Good 020 R4	good		St: 001 020 R2	NVM fail
	St: 002 020 R2	bad clock		St: 040 020 R2	Inter processor communication failure
	St: 010 020 R2	low battery		St: 056 020 R2	all status flags are set
	St: 008 020 R2	switch weld		St: 040 020 R2	Inter processor communication failure
	St: 056 020 R2	All status flags are set		St: 020 020 R2	ETBC Open
16	BATTERY STATUS				
	bAt :Good 023 R4	good		bAt :bAd 023 R4	bad
	bAt :Lo 023 R4	low		bAt : 1000 024 R4	on hours
17	POWER FACTOR AND FREQUENCY				
	PF 1.000 Hz 031 R4	power factor		50.000 Hz 031 R4	frequency
18	VOLTAGE DISPLAYS				
	P1 240.0 V 032 R4	phase 1		P2 226.8 V 033 R4	phase 2

	P3 229.7 V 033 R4	phase 3		3P 240.0 V 034 R4	maximum voltage
	P1 240.0 V 035 R4	maximum voltage (phase 1)		P2 275.0 V 035 R4	maximum voltage (phase 2)
	P3 275.0 V 035 R4	maximum voltage (phase 3)		3P 195.0 V 036 R4	minimum voltage
	P1 184.6 V 037 R4	minimum voltage (phase 1)		P2 187.6 V 037 R4	minimum voltage (phase 2)
	P3 188.5 V 037 R4	minimum voltage (phase 3)		23:12:16 t V 038 R2	maximum voltage time
	23:12:16 t V 039 R2	maximum voltage time (phase 1)		23:12:16 t V 041 R2	maximum voltage time (phase 2)
	04.07.12 d V 042 R2	maximum voltage date (phase 3)		04.07.12 d V 043 R2	maximum voltage date
	04.07.12 d V 044 R2	maximum voltage date (phase 1)		04.07.12 d V 045 R2	maximum voltage date (phase 2)
	04.07.12 d V 046 R4	maximum voltage date (phase 3)		23:12:35 t V 047 R2	minimum voltage time
	23:12:35 t V 048 R2	minimum voltage time (phase 1)		23:12:35 t V 049 R2	minimum voltage time (phase 2)
	23:12:35 t V 050 R2	minimum voltage time (phase 3)		05.07.12 d V 050 R2	minimum voltage date
	05.07.12 d V 051 R2	minimum voltage date (phase 1)		05.07.12 d V 052 R2	minimum voltage date (phase 2)
	05.07.12 d V 053 R2	minimum voltage date (phase 3)			
NETWORK STATUS AND CONTROL					
19	LAN COMMUNICATION				
	LAn :000 053 R0	Not available		LAn :001 053 R0	Access disabled
	LAn :002 053 R0	Initialized		LAn :016 053 R0	Found
	LAn :032 053 R0	Connected			
Note – For codes 00 to 15 the LAN status annunciator does not glow, for codes 16 to 31 the LAN status annunciator will flash at approximately 2 Hz and for codes 32 to 47 this annunciator glows continuously.					
20	MISCELLANEOUS DISPLAYS				

	07:54:55 t 070 R4	Time of loss of voltage / voltage fail) of phase 1		05:05:12 d 071 R4	Date of loss of voltage of phase 1
	07:34:55 t 072 R4	Time of loss of voltage of phase 2		05:05:12 d 073 R4	Date of loss of voltage of phase 2
	07:34:00 t 074 R4	Time of loss of voltage of phase 3		05:05:12 d 075 R4	Date of loss of voltage of phase 3
	10 076 R4	Power off hour for less than 1 minute		125 077 R4	Power on-off count
	on000125 078 R4	Cumulative Power ON Minute		oF000188 079 R4	Cumulative Power OFF Minute
	oP1 on 080 R4	Instantaneous state of output 1		oP2 oF 081 R4	Instantaneous state of output 2
	Ch1 : 1000 086 R4	Energy pulsing configuration of output 1 (channel number and pulses/KWh)		Ch2 : 10 087 R4	Energy pulsing configuration of output 2 (channel number and pulses/KWh)
	L3: 150 100 R4	Current angle (deg), relative to voltage on same phase: phase 1		L2: 312.52 099 R4	Current angle (deg), relative to voltage on same phase: phase 2
	L3: 150 100 R4	Current angle (deg), relative to voltage on same phase: phase 3		n-AL 089 R4	Cover status (not allowed/un-authorized)
	AL 088 R4	Cover status (allowed/authorized)		tar 01 017 R2	Active configuration number
	100 052 R4	Max number activations allowed (by button push)		054 053 R4	Number of activations remaining (by button push)
	122 067 R4	Number of Aux - 2 Connect/disconnect switch cycles			
21	ENGINEERING MODE DISPLAY				

	EnG	Engineering mode		LEdtSt	LED Test
	Co-tSt 085	Contactor Test		PhCnF	Phase Configuration
	Md-rSt	MD Reset		tSt-Cd	Test Mode
	bILL	Ask Conformance for Bill		SoLAr001 098	Configuration named “SOLAR001” stored at location x
	ACCEPt	Request is Accepted		rEJECt	Request is rejected
22	TOTAL HARMONIC DISTORTION				
	P1 00.50 V 101	Phase 1 Voltage total Harmonic distortion in %		P1 00.50 A 104	Phase 1 Current total Harmonic distortion in %
	P2 00.50 V 102	Phase 2 Voltage total Harmonic distortion in %		P2 00.50 A 105	Phase 2 Current total Harmonic distortion in %
	P3 00.50 V 103	Phase 3 Voltage total Harmonic distortion in %		P3 00.50 A 106	Phase 3 Current total Harmonic distortion in %
23	PHASE SEQUENCE				
	SEQ ok V 107	Phase Sequence Ok		SEQ Inv V 107	Invalid Phase Sequence (Wiring Errors)
24	CONTACTOR TEST (LC & RELAYS)				
	rL2-On	Relay 1 On		rL2-OFF	Relay 1 Off

Note:

Some of the displays are build specific and may not be available in all builds.

Appendix 2. DLMS OBIS Codes

All meter objects are read through communication interfaces. For example the meter clock, energy registers, load survey profiles, events etc. The objects in Premier 210 are represented by logical attributes in conformance with DLMS/COSEM identification model. This model is known as Object Identification System (OBIS) and uses six value groups in a hierarchical structure to enable different manufacturers use same logical names and data collectors to read data from different meters through a supported data collecting application.

The following table lists the OBIS codes for various Parameters in Premier 210 meters to provide a logical reference in conformance with DLMS/COSEM object model.

Meter Identification Parameters	**Class ID**	**OBIS code**
Meter Serial Number	1	0.0.96.1.0.255
Manufacturer Serial Number	1	0.0.96.1.1.255
Utility Serial Number	1	0.0.96.1.2.255
Meter Model	1	0.0.96.1.3.255
Meter Payment Type	1	0.0.96.1.4.255
Meter Phase Type	1	1.0.0.2.4.255
Firmware Version	1	1.0.0.2.0.255
Transformer Ratio- Current (Num)	1	1.0.0.4.2.255
Transformer Ratio-Voltage (Num)	1	1.0.0.4.3.255
Transformer Ratio- Current (Den)	1	1.0.0.4.5.255
Transformer Ratio-Voltage (Den)	1	1.0.0.4.6.255

Energy and Independent Parameters	**Class ID**	**OBIS code**
Cumulative Billing Count	1	1.0.0.1.0.255
Active energy, export, net	3	1.0.2.8.0.255
Active energy, import, net	3	1.0.1.8.0.255
Reactive energy, export	3	1.0.4.8.0.255
Reactive energy, import	3	1.0.3.8.0.255
Reactive energy, lag+lead while active export	3	1.0.132.8.0.255
Reactive energy, lag+lead while active import	3	1.0.131.8.0.255
Apparent energy, type 1, all quadrants	3	1.0.138.8.0.255

Apparent energy, type 2, while active import	3	1.0.9.8.0.255
Apparent energy, type 6, while active export	3	1.0.10.8.0.255
Reactive energy Lag (Q1 + Q3)	3	1.0.141.8.0.255
Reactive energy Lead (Q2 + Q4)	3	1.0.142.8.0.255
Active energy (kWh) Import –Phase1	3	1.0.21.8.0.255
Active energy (kWh) Import –Phase 2	3	1.0.41.8.0.255
Active energy (kWh) Import –Phase 3	3	1.0.61.8.0.255
Active energy (kWh) Export – Phase1	3	1.0.22.8.0.255
Active energy (kWh) Export – Phase2	3	1.0.42.8.0.255
Active energy (kWh) Export – Phase3	3	1.0.62.8.0.255
Active energy(kWh) imp + Exp.(Forwarded) - 3Phase	3	1.0.15.8.0.255
Reactive energy (kVARh) Imp.+Exp.(Forwarded)– 3 Phase	3	1.0.143.8.0.255
Reactive energy (kVARh) Imp. 'minus' Exp.(Net) – 3 phase	3	1.0.140.8.0.255
Reactive energy, import while active import	3	1.0.5.8.0.255
Reactive energy, export while active import	3	1.0.8.8.0.255
Reactive energy, import while active export	3	1.0.6.8.0.255
Reactive energy, export while active export	3	1.0.7.8.0.255
Apparent energy, type 10, while reactive import	3	1.0.138.8.0.255
Apparent energy, type 11, while reactive export	3	1.0.138.8.0.255
MAC ID	43	0.1.25.2.0.255

Appendix 3. Events supported in Premier 210

Premier 210 has dedicated data logs for different event types. The event logs are typically used to supervise: the status of the meters, the status of the electricity distribution network (over/under voltages, currents), power failure/restoration, frauds or tampers, and the status of contactors (if used).

The following table lists the OBIS codes for each Event Log in Premier 210 meters to provide a logical reference in conformance with DLMS/COSEM object model.

Sr. No.	Name	Parameters	OBIS Code
Scalar Unit Profile for all Events			***1.0.98.128.7.255***
1.	Standard Log	***Standard Log Value Profile***	***0.0.99.98.0.255***
		Event time stamp	0.0.1.0.0.255
		Standard event top record number	1.65.94.65.100.255
		Event code	0.0.96.11.0.255
2.	Power Quality Event Log	***Power Quality Log Value Profile***	***0.0.99.98.1.255***
		Event time stamp	0.0.1.0.0.255
		Power Quality event top record number	1.65.94.65.101.255
		Event code	0.0.96.11.1.255
		Magnitude	1.65.94. 65. 102.255
		Duration	1.65.94. 65. 103.255
3.	Power Failure Event Log	***Power Failure Log Value Profile***	***0.0.99.98.2.255***
		Event time stamp	0.0.1.0.0.255
		Power failure event top record number	1.65.94.65.104.255
		Event code	0.0.96.11.2.255
		Duration	1.65.94. 65. 105.255
4.	Fraud Detection Event Log	***Fraud Detection Log Value Profile***	***0.0.99.98.3.255***
		Event time stamp	0.0.1.0.0.255
		Fraud event top record number	1.65.94.65.107.255
		Event code	0.0.96.11.3.255
		Duration	1.65.94. 65. 106.255

Sr. No.	Name	Parameters	OBIS Code
5.	Switch Event Log	***Switch Event Log Value Profile***	***0.0.99.98.4.255***
		Event time stamp	0.0.1.0.0.255
		event top record number	0.0.96.15.4.255
		Event code	0.0.96.11.4.255
6.	Current Event Log	Current Event Log Value	0.0.99.98.5.255
		Event time stamp	0.0.1.0.0.255
		Event top record number	1.65.94.65.108.255
		Event code	0.0.96.11.5.255
		Magnitude	1.65.94. 65. 110.255
		Duration	1.65.94. 65. 111.255
7.	Duration Based Events	Duration based event log value	0.0.99.98.6.255
		Event time stamp	0.0.1.0.0.255
		Event top record number	1.65.94.65.108.255
		Event code	0.0.96.11.5.255
		Duration	1.65.94. 65. 111.255

A list of events supported in Premier 210 is tabulated below:

Event Code	Event Name	Log ID
3	Low Battery	1
4	Meter Hw fault/Critical Error	1
7	Ready for Connection	5
10	Time Synchronization	1
11	Power Supply Failure(last gasp)	3
12	Power Supply Restored	3
13	L1 Under Voltage	2
14	L1 Over Voltage	2
15	Wrong Phase Sequence	1
16	Unauthorized Export of Energy	1
17	Local Read	1
18	Local Change of Parameter	1

Event Code	Event Name	Log ID
19	Firmware Activation	1
20	Password Changed	1
21	L2 Under Voltage	2
22	L2 Over Voltage	2
23	L3 Under Voltage	2
24	L3 Over Voltage	2
25	L1 Current Reversal	1
26	L2 Current Reversal	1
27	L3 Current Reversal	1
28	L1 Voltage Missing/failure	3
29	L2 Voltage Missing/failure	3
30	L3 Voltage Missing/failure	3
31	Terminal Cover Removed	4
32	Terminal Cover Replaced	4
33 to 100	Unused for future Use	
128	Device ID Change	1
129	Key Change	1
130	MD reset	1
131	Inter processor communication fail	1
132	Front Cover Removed	4
133	Front Cover Replaced	4
135	Tariff Invocation	4
136	Tariff Download	1
137	Time Set	1
138	Switch Weld	1
139	L1 Under Voltage Restoration	2
140	L2Under Voltage Restoration	2
141	L3 Under Voltage Restoration	2
142	L1 Over Voltage Restoration	2

Event Code	Event Name	Log ID
143	L2 Over Voltage Restoration	2
144	L3 Over Voltage Restoration	2
145	L1 Current Reversal Restoration	1
146	L2 Current Reversal Restoration	1
147	L3 Current Reversal Restoration	1
148	L1 Voltage Missing/failure Restoration	3
149	L2 Voltage Missing/failure Restoration	3
150	L3 Voltage Missing/failure Restoration	3
152	Wrong Phase Sequence Restoration	1
154	Program Re-boot	1
155	Switch Weld Restoration	1
156	Unauthorized Export of Energy End	1
157	Deviation of Local to UTC changed	1
158	NVM Failure	1
159	Authorised Cover Open Time Start	1
160	Authorised Cover Open Time End	1
161	Authorised Cover Open Time Set	1
162	Low Power Factor Start Phase 1	1
163	Low Power Factor Start Phase 2	1
164	Low Power Factor Start Phase 3	1
165	Low Power Factor End Phase 1	1
166	Low Power Factor End Phase 2	1
167	Low Power Factor End Phase 3	1
168	Average Low Power Factor Start	1
169	Average Low Power Factor End	1
170	Over Current Start on Phase1	1
171	Over Current Start on Phase2	1
172	Over Current Start on Phase3	1
173	Over Current End on Phase1	1

Event Code	Event Name	Log ID
174	Over Current End on Phase2	1
175	Over Current End on Phase3	1
176	Voltage Sag Start on Phase1	2
177	Voltage Sag Start on Phase2	2
178	Voltage Sag Start on Phase3	2
179	Voltage Sag End on Phase1	2
180	Voltage Sag End on Phase2	2
181	Voltage Sag End on Phase3	2
182	Voltage Swell Start on Phase1	2
183	Voltage Swell Start on Phase2	2
184	Voltage Swell Start on Phase3	2
185	Voltage Swell End on Phase1	2
186	Voltage Swell End on Phase2	2
187	Voltage Swell End on Phase3	2
203	Start Under Frequency	5
204	End Under Frequency	5
221	Enable Demand Limit for Aux2	5
222	Disable Demand Limit for Aux2	5
232	Aux2 On by Priority Override	5
233	Aux2 Off by Priority Override	5
234	Aux2 off Demand Limit	5
249	Reset Register	1
250	Change Mac ID	1
259	Inter processor communication restore	1
260	NIC antenna switched to internal	1
261	NIC antenna switched to external	1

Appendix 4. List of Display Flags and Corresponding Meter States

Changes in the meter state as a result of event occurrences /varying conditions are indicated by flags on the meter display. Some flags warrant attention. Therefore, when a flag is set, the service icon glows. This is an indication that due service by authorised personnel is needed for maintenance.

A list of possible meter flags and their descriptions is tabulated below:

Flag	Description
St : 01	Indicates NVM failure
St : 02	Indicates Real time clock error
St :10	Indicates Low battery
St : 03	Indicates NVM failure and error in time clock
St : 11	Indicates NVM failure and Indicates Low battery
St:12	Indicates error in time clock and Low battery
St: 13	Indicates NVM failure, error in time clock, and Low battery
St: 08	Switch Weld
St:20	ETBC Open
St:40	Inter processor communication failure

NOTE:

The **St: Good** flag on the display indicates that meter is in a healthy state.

Appendix 5. List of Instantaneous LS Parameters

Load Survey profile data can be captured for any 16 out of the following 24 parameters by the secondary LS logger in Premier 210.

Instantaneous Parameters	Class ID	OBIS code
Average Voltage Phase 1	3	1.0.32.4.0.255
Average Voltage Phase 2	3	1.0.52.4.0.255
Average Voltage Phase 3	3	1.0.72.4.0.255
Average Line Current Phase 1	3	1.0.31.4.0.255
Average Line Current Phase 2	3	1.0.51.4.0.255
Average Line Current Phase 3	3	1.0.71.4.0.255
Average Mean Voltage	3	1.0.12.132.0.255
Average Mean Line Current	3	1.0.11.132.0.255
Average Total Disharmonic Distortion in Current Phase 1	3	1.0.31.24.124.255
Average Total Disharmonic Distortion in Current Phase 2	3	1.0.51.24.124.255
Average Total Disharmonic Distortion in Current Phase 3	3	1.0.71.24.124.255
Average Total Disharmonic Distortion in Voltage Phase 1	3	1.0.32.24.124.255
Average Total Disharmonic Distortion in Voltage Phase 2	3	1.0.52.24.124.255
Average Total Disharmonic Distortion in Voltage Phase 3	3	1.0.72.24.124.255
Average Meter Temperature	3	1.0.149.4.0.255
Average Active Power , Signed	3	1.0.1.4.0.255
Average Reactive Power , Signed	3	1.0.3.4.0.255
Maximum Active Power , Signed	3	1.0.1.6.0.255
Maximum Reactive Power , Signed	3	1.0.3.6.0.255

Minimum Active Power , Signed	3	1.0.1.3.0.255
Minimum Reactive Power , Signed	3	1.0.3.3.0.255
Instantaneous Active Power , Signed	3	1.0.1.7.0.255
Instantaneous Reactive Power , Signed	3	1.0.3.7.0.255
Voltage Imbalance	3	1.0.12.178.0.255
Temperature	3	1.0.149.74.0.255

Appendix 6. Phase Icon Indicators

The phase indicators show the presence of phase voltages in the meter with the characters **1**, **2**, and **3** representing **A**, **B**, and **C** phases respectively. For meters on two phase voltages, indicators glow only for the configured phase combination such as **2, 3** representing **B** and **C** or **1, 2** representing **A** and **B** and so on. The respective icons will glow only if the phase voltage of the configured phase is greater than 75% of nominal voltage.

In addition, the respective phase indicators will continuously flash, if the Line current of a configured phase is greater than 0.5% of I Basic Current and current direction is reverse on that phase.

NOTE:

The same icons indicate presence of phase voltages and reverse run conditions. But in case of reverse run conditions the phase icons flash continuously. This indication takes precedence over the Phase presence. The indicators also flash if the meter has been configured for Co-Generation.

Notes

Notes

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